

Chapter 121 — Urinary Tract Fistulae

- **Definition:** a urinary tract fistula is an **abnormal epithelial- or mesothelial-lined connection** between the urinary tract (kidney, ureter, bladder, urethra) and another body cavity (gastrointestinal, vascular, reproductive) or the skin (cutaneous).
- **Causes are myriad** — iatrogenic, malignancy, inflammation, infection, ischaemia, trauma and parturition. **In the developed world most fistulae are iatrogenic**, the sequel of a surgical intervention.
- **Presentation varies with the anatomical connection and the cause**, so diagnosis can be difficult; a **high index of suspicion and a systematic approach** are required to confirm the diagnosis and begin management.
- **The burden is physical, psychosocial, medicolegal and financial.** Diagnosis should be expeditious without unnecessary testing, with open and honest dialogue. Until definitive therapy, **timely management of leakage, infection and skin breakdown** improves quality of life (QoL) and preserves the patient-provider relationship.
- **The principles of management are virtually universal to all fistula types** (Table 121.1), although they rest on very limited evidence.
- **Prevention is paramount** because most fistulae in industrialised countries are iatrogenic: techniques to avoid injury, vigilant intraoperative evaluation for injury, and techniques that enhance detection and repair of injury when found.
- **Once a fistula is diagnosed**, optimise healing conditions — improve nutrition and eliminate infection (thereby minimising inflammation).
- **Malignancy causes fistulae** both directly and through the sequelae of treatment (for example, radiation); **anyone with a history of, or suspicion of, malignancy needs a work-up including biopsy of the fistula site.**
- **Continuous urinary drainage, diversion of urine away from the tract and relief of obstruction are crucial.** A continuous flow of urine maintains fistula patency and allows epithelialisation of the tract, which reduces the chance of spontaneous closure.
- **A minority heal conservatively; the vast majority need surgery.** The surgical principles (Table 121.2) are independent of fistula location and surgical route, and are likewise not supported by robust evidence.
- **Factors determining approach and success:** size and number of fistulae, complexity, location relative to trigone and ureters, need for concurrent procedures, tissue quality and need for interposition, accessibility, comorbidity, prior radiation therapy and surgeon experience.
- **The first repair has the highest chance of success** — hence optimisation of patient- and surgeon-related factors before the first attempt.

TABLE 121.1 PRINCIPLES OF FISTULA MANAGEMENT

Principle	Practical meaning
Optimise nutritional status	Healing capacity
Control and eradicate infection	Minimises inflammation
Establish continuous urinary drainage	Catheter and/or stent
Eliminate any obstruction	By removal or bypass
Consider malignancy	Primary or recurrent — always biopsy

TABLE 121.2 PRINCIPLES OF SURGICAL REPAIR OF URINARY TRACT FISTULAE

Phase	Principles
Before surgery	Evaluate for concomitant injury ahead of time; rule out malignancy
Exposure and preparation	Adequate exposure; remove all foreign material
Tissue handling	Mobilise surrounding tissue for a tension-free closure; ensure vascularised orifice edges
Closure	Multiple layers; avoid overlapping suture lines; watertight
Reinforcement	Strongly consider tissue interposition
After closure	Establish proper urinary drainage

- **What the tables are for:** Table 121.1 governs the whole course of management; Table 121.2 governs the operation itself, whatever the route.
- **Pattern:** every item removes a reason for the tract to persist — flow of urine, infection, obstruction, foreign body, tension, ischaemia, apposed suture lines, or unrecognised tumour.
- **Trap:** forgetting biopsy — a fistula may be the only sign of recurrent malignancy.

Concept — *a fistula persists because urine keeps flowing through a tract that cannot heal; every management principle either diverts that flow or restores the tissue's capacity to heal.*

UROGYNÆCOLOGIC FISTULAE

Vesicovaginal Fistula

Aetiopathogenesis and Epidemiology

- **Global burden:** an estimated 0.5–2 million females live with untreated urogenital fistulae; true prevalence is unknown because of limited access to care and documentation.
- **Vesicovaginal fistula (VVF)** — abnormal bladder–vagina connection — is the **commonest acquired urinary tract fistula, >75% of genitourinary (GU) fistulae**.
- **History:** the earliest evidence is a large VVF in the mummified Queen Henhenit, wife of King Menthotep II (about 2050 BCE).

VVF AETIOLOGY — DEVELOPING VERSUS DEVELOPED WORLD

	Developing world	Developed world
Dominant cause	Obstetric (>90%) — obstructed labour, inadequate care	Iatrogenic, unrecognised injury — >75% after hysterectomy
Where	Sub-Saharan Africa and South Asia; 30,000–130,000 new cases a year	Obstetric VVF extremely rare, usually after instrumental delivery
Mechanism	Fetal head impacted in the pelvis for days (3–5 days of labour) → pressure → ischaemia → necrosis → large devascularised defect	Blunt or sharp dissection, electrocautery thermal injury, clamp or retractor crush, or posterior bladder wall caught in the vaginal cuff closure
Extent	Level of arrest determines injury; may involve trigone, urethra, ureter	Usually solitary, supratrigonal, at the cuff

- **Obstructed labour injury complex:** multiorgan damage — renal, urinary, reproductive, anorectal, orthopaedic and neuromuscular — with profound psychosocial suffering. Contributors: **(1) young age with a small immature bony pelvis, (2) poor nutrition, (3) geographic isolation from operative obstetric care.**
- **Common final pathway in iatrogenic VVF:** devascularisation → ischaemia → necrosis → breakdown; urine then flows continuously from the higher-pressure bladder into the lower-pressure vagina or abdomen, **keeping the tract patent and allowing epithelialisation.**

Hysterectomy and urological injury

- **Scale:** about 498,000 hysterectomies a year in the United States; about one-fifth of females report prior hysterectomy.
- **Injury rates:** historically **bladder 1–5%** and **ureter 0.2–2.5%** of hysterectomies. Contemporary meta-analyses: urological injury after laparoscopic or robotic hysterectomy is **<1%** and does not differ from vaginal hysterectomy.
- **Risk raised by anatomical distortion:** intraoperative bleeding, prior pelvic surgery, endometriosis, fibroids, adnexal masses, prior radiation and malignancy; and by **concomitant prolapse repair, concomitant incontinence procedure, or mesh for prolapse** — so VVF risk tracks surgical complexity.
- **VVF after hysterectomy:** overall about **0.8 per 1000**; lowest after vaginal (about 0.2/1000), highest after laparoscopic (about 2/1000). A national database found the highest 30-day fistula rate after laparoscopically assisted vaginal hysterectomy (LAVH) and the lowest after transvaginal hysterectomy.
- **Radical hysterectomy for cervical cancer:** VVF more frequent with the laparoscopic than the open route.
- **Most postoperative fistulae arise from unrecognised injury. Immediate recognition and repair lowers VVF rates** compared with delayed recognition, and only a small minority of fistulae follow a recognised injury.
- **Litigation:** urological injury carries a **91-fold higher risk of litigation**.

Universal cystoscopy at hysterectomy

- **For:** high detection rates in routine-cystoscopy series; an institutional universal-cystoscopy policy reduced delayed urological injuries, doubled intraoperative detection and eliminated VVF.
- **Against:** a large systematic review found intraoperative detection rose **fivefold with no fall in delayed injury**; a population study found no difference in 30-day delayed lower-tract injury; cystoscopy adds operative time (median about 17 minutes) and raises urinary tract infection (UTI) risk.
- **Cystoscopy does not replace careful retroperitoneal dissection and constant awareness of the bladder and ureteric course.** Methylene blue instillation may also help.

SOCIETY RECOMMENDATIONS — CYSTOSCOPY AT HYSTERECTOMY

Body	Recommendation
American College of Obstetricians and Gynecologists (ACOG)	No recommendation after benign hysterectomy without pelvic reconstruction
Society of Obstetricians and Gynaecologists of Canada (SOGC)	Low threshold , individualised
American Urogynecologic Society (AUGS)	Routine with any pelvic reconstructive procedure (prolapse or anti-incontinence)
American Association of Gynecologic Laparoscopists (AAGL)	Routine after any laparoscopic or robotic hysterectomy

Other causes

- **Pelvic reconstructive surgery, especially with synthetic mesh:** the foreign body sustains inflammation that prevents healing and maintains a tract; infection, particularly of **non-type 1 mesh**, may contribute.
- **Radiation** (cervical, endometrial, colorectal, bladder cancer): VVF rates up to 13% in high-resource versus 0.2% in low-resource countries. **Mechanism: fibrosis of the lamina propria and obliterative arteritis of small and medium arteries** → atrophy or necrosis → urothelial ulceration → fistula.
 - **0.6–2% of females irradiated for cervical cancer** develop VVF.
 - Typical presentation **1.5–2 years after treatment**, but **up to 30 years** later.
 - **Dose dependent; higher in smokers.**

- **Bevacizumab** (anti-vascular endothelial growth factor [VEGF] monoclonal antibody for advanced cervical, ovarian and colorectal cancer) raises fistula risk through its hypoxic effects.
- **Primary and recurrent malignancy itself** causes fistula — **every VVF in a female with prior malignancy must be evaluated for recurrence.**
- **Rare causes:** chronic infection, congenital, vaginal lasers, endometriosis, foreign bodies (pessaries, intrauterine devices).

Concept — *obstetric and iatrogenic VVF share one mechanism — ischaemic necrosis followed by a continuous pressure-driven flow of urine — but differ in scale: prolonged impaction produces a large devascularised field, whereas a surgical injury produces a small focal defect at the cuff.*

Classifications of Vesicovaginal Fistula

- **No universal system:** at least 25 exist, almost all for obstetric fistulae. The commonest are **Goh** and **Waldijk** (Table 121.3); the World Health Organization (WHO) also classifies fistulae as simple or complex.
- **Both address urethral involvement and size.** **Waldijk** is built on the **closing (continence) mechanism** and extent of urethral involvement; **Goh** uses a **fixed point — the external urethral meatus** — to describe location and, combined with size, the extent of urethral loss.

TABLE 121.3 GOH AND WALDIJK CLASSIFICATIONS OF VESICOVAGINAL FISTULA

Axis	Goh	Waldijk
Reference point	Distal edge from external urethral meatus	Closing mechanism and urethral involvement
Type	1: >3.5 cm · 2: 2.5–3.5 cm · 3: 1.5–2.5 cm · 4: <1.5 cm	I: closing mechanism spared · IIA: no (sub)total urethral involvement · IIB: (sub)total urethral involvement · III: miscellaneous (ureteric, exceptional)
Subtype	—	A: without circumferential defect · B: with circumferential defect
Size	A: <1.5 cm · B: 1.5–3 cm · C: >3 cm	Small <2 cm · medium 2–3 cm · large 4–5 cm · extensive ≥6 cm
Tissue	i: none or mild fibrosis, vagina ≥6 cm, normal capacity · ii: moderate or severe fibrosis, reduced vaginal length or capacity · iii: special (post-radiation, ureteric, circumferential, previous repair)	—
Prognosis	Worse with higher type	Closure falls and incontinence rises with higher type

- **Pattern:** both systems grade **proximity to, or involvement of, the continence mechanism** — the closer the fistula to the meatus, the worse closure and continence.
- **Predictive value:** **Goh predicts surgical closure better than Waldijk** in comparative studies, and proximity to the meatus predicts closure and continence after repeat repair. In the developed world, Goh type did not predict closure but **continence worsened with higher type.**
- **Trap:** Goh type 1 is the **most distant (best)** and type 4 the **closest (worst)** — the numbering runs towards the meatus.
- **Practical bottom line:** classifications serve research and communication; every VVF is unique. **Always describe size, location (trigonal versus supratrighal), tissue loss, ureteric and urethral involvement** for planning.

Concept — *fistula classification is really a measure of threat to continence: distance from the meatus and involvement of the closing mechanism predict the patient who is closed but still wet.*

Diagnosis of Vesicovaginal Fistula

- **High index of suspicion above all**, then history and examination; office tests, endoscopy and imaging confirm an uncertain diagnosis and evaluate adjacent structures.

Clinical presentation.

- **Commonest complaint: urinary leakage.** Characterise viscosity, colour, odour, volume, onset relative to events, provoking and relieving factors. **Coexisting mechanisms** — stress urinary incontinence (SUI), overactive bladder (OAB), mixed urinary incontinence (MUI), overflow, post-void dribble — may coexist.
- **Classic VVF: constant leakage day and night, regardless of position or activity, often without urge.** A large tract may abolish the desire to void because the bladder never fills. A small VVF may cause only positional or minimal leakage.
- Also: dysuria, urgency, frequency, **recurrent or persistent UTI**, gross haematuria or blood-tinged discharge; patients may mistake leakage for vaginal discharge. **Pain is uncommon** except with skin irritation or radiation change.
- **Red flags** — fever, chills, malaise, nausea, vomiting, flank pain, ileus: suggest **urinoma and/or ureteric obstruction** → evaluate for concurrent ureteric injury and manage immediately.
- **Timing points to mechanism: unrecognised or failed cystotomy repair presents within the first few days; thermal or crush injury presents at 7–10 days;** radiation fistula presents years later; **VVF may be the only sign of recurrence** — biopsy the tract in anyone with prior malignancy.
- **Misdirected treatment is common:** patients may already have been treated for SUI, OAB or MUI without benefit.
- **Obstetric history:** in industrialised countries, obstetric VVF follows **caesarean and forceps delivery** (unrecognised bladder injury), leaking within days. **Exclude overflow** from neurological or stretch injury after epidural bladder mismanagement.
- **Obtain and review prior operative notes.** Gynaecological history: endometriosis, abnormal bleeding, intrauterine device, pelvic inflammatory disease, pessary, indwelling catheter.
- **Optimise fitness:** smoking cessation, glycaemic control, nutrition, and a support system matching physical and psychosocial needs.

Examination findings.

- **Abdomen:** distension, tenderness, costovertebral angle tenderness; fresh incisions for infection, leakage or dehiscence; confirms prior surgery in poor historians.
- **Perineum:** skin irritation from persistent moisture.
- **Speculum:** cervix and **cuff integrity**; pooled fluid for **creatinine** — **fluid creatinine approximating urine creatinine confirms urine** (soaked pads or tampons can also be tested). Note number, size and location of orifices. **The vaginal apex is the commonest site of post-hysterectomy VVF.**
- **If no defect is seen:** backfill the bladder with methylene blue and watch; if still no leak, pack the vagina with gauze or tampon and walk for **15–60 minutes** — **blue at the apex indicates a tract.** Granulation or friable tissue may signal the tract.
- **Prior malignancy or radiation: biopsy the defect or any friable mass.** Severe post-radiation stenosis may limit examination → **examination under anaesthesia** with endoscopy and imaging of ureteric integrity.
- **Vaginal atrophy:** offer **vaginal oestrogen** before repair. Examination also judges tissue quality, feasibility of repair and the likely route (vaginal versus abdominal).

Investigations

Dye tests.

- **Simple office confirmation of a connection**; small series show high sensitivity.
- **Single dye test**: backfill with **methylene blue or indigo carmine (about 300 mL)** → vaginal examination for blue leak → if unseen, tampon and ambulate → inspect.
- **Stain position on the tampon localises: proximal stain → VVF; distal stain → urethrovaginal fistula or SUI**. Leak through the cervical os indicates vesicouterine fistula.
- **Double dye test** distinguishes VVF from ureterovaginal fistula: oral **phenazopyridine 100–200 mg** → after 1 hour (urine orange) → backfill blue → tampon → ambulate.

DOUBLE DYE TEST — READING THE TAMPON

Stain	Source of dye	Interpretation
Orange	Kidney (oral phenazopyridine)	Ureterovaginal fistula
Blue	Bladder (instilled dye)	VVF
Both	Both	Coexisting VVF and ureterovaginal fistula

- **Trap**: the double dye test **does not replace upper tract imaging**, which is imperative whenever ureterovaginal fistula is suspected.

Endoscopic tests.

- **Cystoscopy is integral** (sensitivity for GU fistula about 92–93%) — for diagnosis and for planning. **Proximity to the ureter, bladder neck involvement or associated mesh** mean major reconstruction; assess capacity and biopsy suspicious masses. Findings decide route and need for ureteric or bladder neck reconstruction, mesh excision, augmentation or diversion.
- **Mature fistulae have smooth epithelialised tracts** and are easier to see; small tracts or a trabeculated bladder make identification hard → probe erythematous patches, granulation or bullous oedema with a **guidewire or 5-Fr open-ended ureteric catheter**; cannulation shows the course.
- **Flexible cystoscopy** (office) allows cystoscopy and vaginoscopy and **retroflexion to detect bladder neck fistulae**. **Cystoscopy under anaesthesia** allows simultaneous retrograde pyelography and stenting — indicated if ureteric obstruction or ureterovaginal fistula is suspected. **Inspect the urethra** for urethrovaginal fistula.
- **Vaginoscopy** shows the apex and the vaginal side of the tract, allows biopsy under vision (vital after malignancy), overcomes radiation stenosis, can be combined with a dye test (including dye at the cervical os for a vesicouterine tract), and evaluates mesh exposure.

Imaging.

- **Cystography** diagnoses VVF when uncertain and shows exact location and number of tracts; **voiding cystourethrogram (VCUG)** shows a concomitant urethrovaginal fistula or rare outlet obstruction (for example, after pelvic trauma).
- **Magnetic resonance imaging (MRI), especially T2-weighted**, helps in difficult cases. Transvaginal colour Doppler ultrasound has been described but is not routine.
- **Lower tract imaging is not mandatory** if the VVF is obvious on examination or endoscopy. **Upper tract imaging is mandatory** to exclude ureteric injury — especially when fistula is suspected but no VVF is found. **Ureteric injury or ureterovaginal fistula coexists in up to 12% of VVF**.
- **Renal ultrasound** screens for hydronephrosis; in obstetric fistula, **absence of grade 3 hydronephrosis has a high negative predictive value** for ureterovaginal fistula.

- **Excretory urography — computed tomography urogram (CTU) with delayed phase** (formerly intravenous pyelogram [IVP]) — whenever upper tract involvement is suspected. **Partial ureteric obstruction with a VVF is a ureterovaginal fistula until proven otherwise.**
- **MRI urography** is reserved for iodinated contrast allergy or renal disease precluding CT.
- **Abnormal or unclear CTU, or suspected obstruction → retrograde pyelogram in theatre with stenting;** if stenting fails → **percutaneous nephrostomy (PCN)**. The nephrostomy tract later permits antegrade ureteroscopy during repair to locate ureters in adhesions and identify the fistulous ureteric orifice.

Urodynamics.

- **Urodynamics (UDS)** rarely help diagnosis but aid planning; about 10–15% of females with fistula also have SUI.
- **Video urodynamics (VUDS)** separate fistula leakage from SUI, urgency urinary incontinence (UUI), MUI, overflow or diverticulum — guiding counselling and concomitant procedures: **fascial sling** for independent SUI; **augmentation or diversion** for radiation-reduced capacity or compliance; **capacity before reimplantation** in case a psoas hitch is needed.
- **Cystogram** distinguishes VVF from other insensible loss — immediate vaginal opacification, **best on the lateral view**; VCUG may show urethrovaginal fistula, stricture or foreign material.
- **Filling a bladder with a large fistula:** pack the vagina, study supine, or plug the fistula with a Foley balloon; for a distal VVF near the bladder neck, a urethral Foley balloon beside the UDS catheter may occlude it → filling cystometry for detrusor overactivity, capacity, compliance and vesicoureteric reflux. The voiding phase remains difficult.

Figures 121.1–121.4: 121.1 cystoscopy — right ureteric orifice beside the fistulous tract; 121.2 urethroscopy — proximal urethral orifice below the bladder neck in a self-catheterising neurogenic female after autologous pubovaginal sling; 121.3 vaginoscopy with blue backfill; 121.4 VCUG — vaginal filling on voiding from a post-hysterectomy VVF.

Concept — *the work-up answers three questions in order: is the fluid urine, where does it come from (bladder, ureter, urethra, uterus), and what else is damaged — and the last question is why upper tract imaging is compulsory even when the VVF is obvious.*

Treatment

General approach.

- **Goal:** eliminate wetness, skin irritation, odour, recurrent infection and occasional pain, following the principles of Table 121.2.
- **Management depends on** size and number, location and proximity to ureters or bladder neck, timing, tissue quality, and surgeon comfort.

Definition of success and prognostic factors.

- **No consensus definition of success. Anatomical closure ignores residual incontinence** — a substantial proportion remain incontinent at 3 months after successful closure.
- **Follow-up is short or absent** (especially in rural Africa) → long-term breakdown and incontinence are underestimated and **success is overestimated**. Future studies should state outcomes explicitly.

Fistula Size

- **Fistulae <3 mm have the greatest chance of spontaneous closure;** size <10 mm favours closure with continuous drainage.

- **Size also predicts surgical failure and residual incontinence:** worse prognosis above about **3 cm (non-irradiated)** and about **2.7 cm (irradiated)** in non-obstetric fistula; size ≥ 1.5 cm predicts residual incontinence and ≥ 3 cm surgical failure in obstetric series.
- **No clear threshold exists**, but size is prognostic for both spontaneous and surgical closure.

Location

- **Most post-hysterectomy VVF are supratrigonal and midline** — approachable vaginally or abdominally.
- **Within 1 cm of a ureter** → **ureteric stenting** for identification and to protect against periureteric oedema in early healing; stents also let the vaginal surgeon **palpate the ureteric course**. Vaginal repairs were associated with a greater VVF–ureter distance than abdominal ones.
- **Ureteric involvement** → **reimplantation** → **abdominal approach**.
- **Bladder neck involvement** → reconstruction for continence, possibly a **pubovaginal sling**.
- **Location dictates interposition:** a **Martius flap is unlikely to reach an apical VVF** (about 8–10 cm from the introitus) → peritoneal or omental flap instead.
- **Location is prognostic:** urethrovesical junction involvement predicts primary failure; distance from the external meatus predicts postoperative incontinence regardless of anatomical closure.

Timing

- **Contentious, with little high-quality data.** The conventional delay lets acute inflammation settle and improves tissue; for obstetric fistula a **≥ 3 -month delay** was recommended (resolution of inflammation, sloughing of necrosis), with an indwelling catheter meanwhile.
- **Counter-argument:** delay increases psychosocial stress without improving outcome.
- **"Early" is not uniformly defined** — days, < 6 weeks or < 3 months. Waaldijk defined early as **< 3 months** and found immediate repair of obstetric fistula achieved very high first-attempt closure. Others describe a difficult window at **4–12 weeks** (immediate < 4 weeks; delayed > 12 weeks).
- **Comparative series:** early and delayed repair give **comparable closure** in most series; one small series found more recurrence with repair < 6 weeks.
- **Minimally invasive early repair is feasible:** laparoscopic repair within 4 weeks and robotic repair of recurrences achieved high closure, though series were non-comparative.
- **European Association of Urology (EAU) consensus:** **no minimum delay before robotic repair**, provided there is **no active infection and no recent irradiation**.

History of Radiation Therapy

- **Radiation VVF presents weeks to years after treatment.** In the largest series, first-attempt transvaginal closure was **under half**; failures were re-repaired after ≥ 6 months.
- **Ideal delay: ≥ 12 months after the end of radiation**, so acute inflammation subsides; meanwhile catheterisation, protective skin ointments and surveillance of the tract for recurrence.
- **Radiation is one of the strongest predictors of repair failure** and brings coexistent dysfunction — **ureteric stricture, poor compliance, intrinsic sphincter deficiency**. Evaluate all lower tract function and **consider urinary diversion** when dysfunction is severe.

TIMING OF REPAIR — THE COMPETING POSITIONS

Setting	Recommended interval	Rationale
Obstetric (traditional)	≥ 3 months	Inflammation settles, necrosis sloughs

Setting	Recommended interval	Rationale
Iatrogenic, clean tissue	Early acceptable (EAU: no minimum delay if no infection or recent radiation)	Comparable closure; less distress
Radiation-induced	≥12 months after radiation ends	Late radiation injury; worst tissue
Some authors	Avoid 4–12 weeks	Active healing phase

Concept — *timing is a trade between tissue quality and patient distress; the trade tilts towards early repair for clean iatrogenic injury and towards long delay only where tissue is still evolving — obstetric necrosis and, above all, radiation.*

Conservative Treatment

Indwelling catheters.

- **Continuous bladder drainage is a treatment in itself**; low-level evidence suggests a **cure rate of about 15%**. Candidacy depends on **size, time since injury and aetiology**; drainage duration and type vary.
- **Size**: no cut-off; spontaneous closure is reported essentially only below 1 cm, and some units trial drainage only below 5 mm (selection bias).
- **Obstetric protocol (Waalwijk)**: only VVF with **necrotic edges** received a catheter trial — **weekly review for 4 weeks**, then surgery if unhealed; **fresh (clean) edges went straight to surgery**. About 15% healed with catheter alone — probably an underestimate, since clean-edged fistulae were never trialled.
- **Time from injury is decisive** because the tract epithelialises (table below); small post-hysterectomy VVF catheterised at once on diagnosis at 10–12 days have all closed.
- **Duration is arbitrary**: 4 weeks (obstetric series), up to 8 weeks (when started <6 weeks from injury). **EAU consensus: if attempted, trial for a minimum of 12 weeks.**
- **Aetiology**: radiation and malignancy fistulae almost never heal with drainage — vascular damage precludes healing.
- **Hypothesis**: indwelling devices may harm reconstruction by inflammation (as shown in male urethral and ureteric reconstruction) — **removing the catheter several days before repair** may improve closure.

CATHETER DRAINAGE — WHO IS LIKELY TO CLOSE

Favourable	Unfavourable
Small (<3 mm best; <1 cm)	Large
Catheter within 3 weeks of injury	Catheter ≥6 weeks after injury (epithelialised)
Fresh surgical or obstetric injury	Radiation or malignancy

Cystoscopic techniques.

- **Electrocoagulation** of the tract with prolonged drainage was first reported by O'Connor and Sokol. Fulguration (transvaginal and/or cystoscopic) plus 2 weeks' drainage succeeded in most fistulae **<3.5 mm**; successes were 0.5–0.6 cm, the failure 0.7 cm. **Because such fistulae may close with drainage alone, the added value of fulguration is uncertain.**
- **Tissue adhesives**: a systematic review reports success comparable to surgery, but it pooled adhesives used as interposition layers as well as plugs.

ENDOSCOPIC SEALANTS FOR VVF

Agent	Technique	Point to know
Fibrin plug (Chevassu technique)	Lyophilised fibrinogen, calcium and thrombin injected via a transinguinal approach; drainage 8 weeks	Plug lysed by fibrinolysis and replaced by connective tissue
Bovine collagen plus fibrin glue	Fulgurate tract → submucosal collagen around tract → fibrin glue plug → vaginal collagen; catheter 3 weeks	Used for a small radiation-induced VVF
Fibrin glue alone	Endoscopic injection	Some needed a second injection
Bovine serum albumin (BioGlue)	Transvaginal injection into tract	Recurrent iatrogenic VVF
Cyanoacrylate	Cystoscopic 0.5 mL, then transvaginal 0.5–1 mL; catheter 8–10 days	Narrow tracts do better (more surface for adherence); failed in the largest tract

- **Complications:** catheter adherent to glue, bladder spasms (anticholinergics), dysuria, haematuria, **stone formation at the glue site** needing laser lithotripsy.
- **Evidence:** only a handful of tiny series → **not recommended by the latest EAU guidelines**. Hypothetically a low-morbidity option for VVF <1 cm in poor surgical candidates.

Surgical Repair of Vesicovaginal Fistula

- **Indication:** VVF persisting after an adequate trial of drainage, or when conservative care is inappropriate.
- **Variables:** size and number; relation to trigone, ureters, bladder neck and urethra; concurrent procedures; tissue quality and need for flaps; access; comorbidity; surgeon experience. **The best results come from the procedure the surgeon is most comfortable with.**
- **Tract excision and interposition** are not proven superior, but are standard reconstructive principles and are of particular value where recurrence risk is high — **irradiated, recurrent, poor tissue**.

Preoperative requirements.

- **Urine culture; treat UTI fully. Biopsy the tract if there is any history of malignancy** (may change the plan). Assess fitness and optimise nutrition.

Preoperative counselling.

- **General risks:** bleeding, infection, venous thromboembolism, injury to adjacent structures.
- **Vaginal routes risk vaginal shortening and dyspareunia**, particularly the **Latzko procedure or colpocleisis** → document baseline sexual function and pre-existing dyspareunia.
- **Prolonged catheterisation**, catheter care and pre-removal studies (for example, cystogram); bladder spasms are self-limiting, and antispasmodics are reasonable.
- **Persistent incontinence despite anatomical closure** is possible; **failure and reoperation** are always possible, although most repairs succeed.

Choosing a surgical route.

- **No ideal approach for all — an optimal approach for each patient**, mostly dictated by surgeon comfort, weighed against the variables in Table 121.4.

TABLE 121.4 CONSIDERATIONS FOR ROUTE OF SURGERY

Consideration	Abdominal (open)	Laparoscopic / robotic	Vaginal
Incision	Large	3–6 ports of 5–12 mm	Hidden

Consideration	Abdominal (open)	Laparoscopic / robotic	Vaginal
Fistula location	Supratrigonal ; narrow vaginal cavity; apex without descensus		Trigonal / distal ; accessible apex
Ureteric proximity	Reimplantation may be needed, or orifice close to fistula (injury risk)		Ureters sufficiently distant; no reimplant concern
Adjuvant procedures	Augmentation ileocystoplasty; reimplantation; diversion; bladder (Tanagho) flap; bladder neck sling		Urethral reconstruction; pubovaginal sling
Sexual function	Normally unchanged		Risk of shortening and dyspareunia
Flaps	Omental; peritoneal; pericolic epiploic appendage; rectus abdominis		Martius ; peritoneal; gracilis myocutaneous
Other indications	Cannot be placed in lithotomy	Cannot be placed in lithotomy; cosmesis and minimally invasive advantages; easier teaching through shared video view	Multiple previous abdominal operations with dense adhesions; no access to general anaesthesia
Perioperative well-being	—	Shorter operative time and stay (than open)	Shorter operative time and stay

- **Pattern:** the open and minimally invasive abdominal columns are almost identical — the **real decision is vaginal versus transabdominal**; laparoscopy or robotics then chooses how the abdomen is entered.

Vaginal VVF repairs — history.

- **Henry van Roonhuysse** (17th century, Dutch): speculum exposure in lithotomy, pared edges approximated with swan-feather quills and silk.
- **J. Marion Sims** first published a systematic, successful vaginal repair using **silver wire**.
- The belief that gynaecologists repair vaginally and urologists abdominally is outdated: in a United States registry, **most vaginal VVF repairs were performed by urologists**.

Transvaginal versus open abdominal repair.

- **No randomised controlled trials (RCTs)** — infeasible and ethically difficult.
- Pooled cohorts and meta-analyses: **similar first-attempt closure (about 90% by either route)**; one pooled analysis suggests slightly better closure transvaginally.
- **Transvaginal advantages:** less blood loss, shorter operation and stay, less pain → some regard it as the **gold standard unless specifically contraindicated**; also **more cost-effective** in a United Kingdom analysis (similar success and reoperation).
- **Transvaginal disadvantages:** vaginal shortening (especially Latzko); poor exposure in a **deep narrow vagina without apical descensus**; complex tracts; **ureteric injury in trigonal or peritrigonal fistulae**; unfamiliarity with cuff anatomy.
- **Abdominal (especially laparoscopic or robotic) advantages:** exposure of the fistula and surrounding structures (for example, the ureter), and shared video view for teaching.
- **The optimal approach is the one in which the surgeon is most confident.**

Transvaginal Repair Techniques

- **Two classic techniques since Sims** — **flap-splitting (layered)** and **Latzko**; both much modified; **no data show either superior**.
- **Flap-splitting** was popularised by **Raz** and colleagues. Key components: **(1) continuous drainage via suprapubic and urethral catheters, (2) adequate exposure, (3) vaginal wall flaps to avoid overlapping suture lines, (4) opportunistic Martius or peritoneal flap**.

TABLE 121.5 FLAP-SPLITTING TECHNIQUE

Step	Sequence
1 Positioning and exposure	Dorsal lithotomy → bimanual examination → prep vagina, perineum, lower abdomen → weighted speculum → cystoscopy to relate VVF to ureteric orifices, stent if close , pass guidewire through tract → optional suprapubic catheter with Lowsley tractor while bladder full → urethral catheter → posterolateral relaxing episiotomy if narrow introitus → ≥8-Fr catheter through the fistula, balloon inflated, gentle traction brings tract into view
2 Incision	Hydrodissect with saline or dilute adrenaline → J- or U-shaped incision , curve circumscribing the tract with a 1–3 mm cuff of epithelium → limbs run towards the apex for distal VVF, towards the introitus for proximal VVF → allows flap advancement beyond the repair (no shortening, no overlapping lines, extra layer)
3 Flap development	Anterior and posterior vaginal flaps raised sharply from the curve, several centimetres in all directions , exposing perivesical fascia → tension-free closure; correct plane limits bleeding and haematoma
4 Tract closure	Tract may be excised → layer 1: 3-0 or 4-0 delayed absorbable , including epithelialised edge, bladder wall and 3–5 mm perivesical tissue; first stitch placed before catheter removal to tag the defect , then balloon deflated and stitch tied; interrupted or running, horizontal or vertical → layer 2: 3-0 or 2-0 imbricating, perpendicular to layer 1 (except very large orifices) → watertight test with 200–300 mL dyed fluid → interposition (Martius or peritoneal) if needed
5 Vaginal wall closure	Trim redundant curve → advance opposite flap ≥2 cm beyond the first two layers (without tension) as the third (or fourth) layer → 2-0 absorbable running, optionally locked → cystoscopy for ureteric efflux, remove stents → urethral catheter unless suprapubic in place (both acceptable) → vaginal pack if bloody
6 Postoperative drainage	The most important part of the repair. Urethral and/or suprapubic catheter for ≥10 days, often ≥2 weeks (optimum never established); two catheters give backup against blockage; cystogram before removal optional

- **Outcomes:** most series report **≥90% first-attempt closure**, including large mixed series with radiation and obstetric cases.
- **Logic of the steps:** tagging stitch, perpendicular layers and flap advancement exist to **prevent overlapping suture lines**; drainage protects the repair from the pressure that created the fistula.

Figures 121.5–121.6 (Ganabathi et al., 1996): a traction Foley in the tract; first layer placed **without excising the tract edges**; perivesical fascia closed with Lembert sutures perpendicular to it.

- **Latzko procedure (1942)** — a **high partial colpocleisis**. Advantages: short operation, minimal blood loss, **no ureteric cannulation or reimplantation** (superficial), rapid recovery. Success **≥90%** in non-irradiated VVF (Table 121.6).

TABLE 121.6 THE LATZKO PROCEDURE

Step	Sequence
1 Positioning and exposure	As for flap-splitting; suprapubic catheter rarely placed
2 De-epithelialisation	Denude vaginal epithelium circumferentially 1–2 cm around the tract , leaving underlying tissue → avoids bladder and ureteric injury
3 Reapproximation of vaginal walls	2-0 or 3-0 absorbable interrupted superficial sutures (not entering bladder) → layer 1 about 0.5–1 cm from the tract on each side, imbricating the defect → layer 2 a further 0.5–1 cm out
4 Additional layers	Martius flap may be added before epithelial closure — denude widely enough to accommodate it

Step	Sequence
5 Epithelial closure	Interrupted or running 2-0 or 3-0 absorbable → cystoscopy, remove stents → urethral catheter unless suprapubic
6 Drainage	Identical to flap-splitting

- **Outcomes:** high cure after the first Latzko in post-hysterectomy series, and failures cured by a second Latzko; **less useful than flap-splitting for obstetric VVF.**
- **Radiation VVF:** both techniques used; the Latzko is favoured for **marked trophic change or proximity to the ureteric orifices** because only the vagina is denuded.
- **Theoretical drawbacks:** no dissection of bladder off vagina (possible tension) and **the bladder orifice is not closed** — yet data do not show higher failure.
- **Regional anaesthesia is possible;** the **downside is vaginal shortening with dyspareunia.**
- **Vaginal cuff excision technique** — principle of **complete scar removal:** circumferential incision 3–5 mm beyond the apical scar → deepened in a **funnel** towards the bladder → all scar and tract excised → an inverted cone of fresh tissue closed in layers. Very high cure in non-irradiated post-hysterectomy VVF; regional anaesthesia, ambulatory, low morbidity.

TRANSVAGINAL TECHNIQUES COMPARED

	Flap-splitting	Latzko	Cuff excision
Principle	Layered closure with offset flaps	High partial colpocleisis	Excise all apical scar
Bladder dissected off vagina	Yes	No	Yes (funnel)
Bladder orifice closed	Yes	No	Yes
Ureteric risk	Present near trigone	Minimal	Present
Interposition	Martius or peritoneal	Martius (wide denuding)	—
Vaginal shortening	Avoided by flap advancement	Main drawback	—
Best niche	Obstetric and most VVF	Post-hysterectomy apex; radiation; near ureters	Post-hysterectomy, non-irradiated

Abdominal Vesicovaginal Fistula Repairs

- **Proposed indications:** large fistula (**≥4 cm primary, ≥2 cm recurrent**), urethral or bladder neck involvement, ureteric involvement, need for ancillary abdominal procedures (augmentation, reimplantation), non-capacious vagina, post-radiation VVF — not universal criteria.
- **Open disadvantages:** large incision — blood loss, infection, longer stay and recovery, more pain, poorer cosmesis.
- **Routes:** intraperitoneal, extraperitoneal or transvesical.
- **Commonest:** the **suprapubic extraperitoneal technique of O'Connor and Sokol (1951)** (Table 121.7).

TABLE 121.7 THE O'CONOR TECHNIQUE

Step	Sequence
1 Positioning	Low lithotomy (vagina and urethra in field) → ureteric catheters pre- or intraoperatively → let the bladder fill without a catheter so it pushes the peritoneum cranially for extraperitoneal entry

Step	Sequence
2 Incision and entry	Low midline → wide extraperitoneal mobilisation of bladder from posterior and lateral attachments (peritoneum often breached) → sagittal midline cystotomy at the dome exposing trigone, orifices and fistula → catheterise ureters if not done
3 Bivalving and excision	Extend cystotomy posteriorly down to the fistula → circumscribe and excise with necrotic tissue → one continuous vertical cystotomy → stay sutures → dissect bladder off vagina ≥2–3 cm beyond the tract (vaginal stent or sponge stick) → trim vaginal edges
4 Vaginal closure ± interposition	One or two layers absorbable → omentum or peritoneum (especially irradiated tissue) fixed 1–2 cm beyond the vaginal closure
5 Bladder closure	Ancillary procedures first (reimplantation, augmentation) → one or two running absorbable layers (mucosa + muscularis; then muscularis + serosa) → suprapubic catheter ± stents → 200–300 mL fill test → optional backup urethral catheter
6 Postoperative	Optional drain (creatinine; haematoma or urinoma) → anticholinergics or β3-agonists for spasm → drainage ≥10–14 days → cystogram

- **Key components:** suprapubic extraperitoneal entry; **bivalving the posterior bladder** to see the VVF and orifices; **incorporating the tract in the cystotomy**; **wide mobilisation** of bladder off anterior vagina.
- **Original series:** single running bladder layer, vaginal closure "**optional and irrelevant**" if the bladder is **well mobilised**, no deliberate avoidance of overlapping lines, no interposition. Later series added two-layer closures and omental or peritoneal flaps.
- **Causes of failure:** carcinoma ± radiation, wound infection, **inadequate bladder mobilisation**.
- Abdominal repair in benign iatrogenic VVF achieves very high pooled success; the procedures are increasingly done laparoscopically or robotically.
- **Suprapubic transvesical approach** (a modification of O'Connor): **limited cystotomy** → tract excised → bladder mobilised circumferentially off vagina → layered closure; **posterior bladder wall flaps** or omentum to avoid overlap. Success similar; the smaller cystotomy may reduce postoperative irritation and voiding dysfunction.
- **Extravesical approach:** no intentional cystotomy; transperitoneal access to the vesicovaginal space **as in post-hysterectomy sacrocolpopexy** → bladder mobilised well beyond tract (a catheter placed through the tract at cystoscopy helps find it) → vagina closed → flap 1–2 cm beyond → bladder in two layers; without a flap, **close bladder and vagina perpendicular to each other**.

ABDOMINAL APPROACHES COMPARED

	O'Connor (bivalving)	Transvesical (limited cystotomy)	Extravesical
Cystotomy	Dome to fistula	Limited	None
Orifices seen	Yes	Yes	No — ureteric injury risk
Interposition	Easy	Limited	Easy
Main advantage	Best identification of the plane and fistula	Less bladder irritation	No large cystotomy (less LUTS, fewer new tracts)
Main drawback	Long suture lines; more LUTS	Restricted mobilisation	Scarred plane — inadvertent cystotomy or colpotomy; infratrighonal fistula hard to reach

Laparoscopic and Robotic Approaches

- **Firsts:** laparoscopic VVF repair 1994; robotic 2005.
- **Open versus minimally invasive transabdominal:** non-randomised comparisons show **equal closure** with **less blood loss, shorter stay and less analgesia** after laparoscopic or robotic repair; operative time similar. Minimally invasive surgery removes morbidity without compromising success.

- **Robotic advantages:** relatively rapid learning curve, three-dimensional view, **7 degrees of freedom**, dual-console teaching, ergonomics.
- **Transvaginal versus minimally invasive abdominal:** the only comparison (a vaginal series against pooled systematic-review data) favoured transvaginal repair for operative time, blood loss and stay.

Concept — *every VVF operation, vaginal or abdominal, is the same reconstruction — separate bladder from vagina, close each without tension, keep the suture lines apart and drain — and the route is chosen by which one best exposes that plane in the individual patient.*

LAPAROSCOPIC AND ROBOTIC VESICOVAGINAL FISTULA REPAIRS

- **Indications mirror open abdominal repair:** ureteric involvement, ancillary abdominal procedures, **narrow vagina with minimal descensus**, surgeon preference.
- **Against:** dense abdominopelvic adhesions or prior inflammation. **Conventional laparoscopy:** limited degrees of freedom make intracorporeal suturing hard for all but advanced laparoscopists, limiting adoption.

Transvesical and O'Connor Approaches

- **Three techniques adapted from open surgery:** transvesical, extravesical and **mixed (O'Connor)**.
- **Distinguish O'Connor from pure transvesical:** in the pure transvesical technique the cystotomy is **not** extended down to the fistula, and **interposition options are much more limited**; the literature often blurs the two.
- **Technique:** posterior-wall cystotomy extended to the VVF (incorporating the defect) → bladder mobilised off vagina **≥2 cm beyond the tract** → vaginal stent (Lucite or an end-to-end anastomosis [EEA] sizer) for **counter-traction** → optional edge excision → vagina closed in one or two layers → bladder in one or two layers (barbed or unbarbed) → interposition of **peritoneum, omentum or sigmoid epiploic appendages**.
- **Closure 85–100% in non-irradiated VVF**, comparable to open and vaginal repair; modifications limit cystotomy size.

Extravesical Approach

- **First described laparoscopically in 1998 (von Theobald).** **No intentional cystotomy** → fewer sutures that could create new tracts or cause lower urinary tract symptoms (LUTS). Steps as for open extravesical repair.
- **Finding the plane:** backfill **150–300 mL** plus vaginal stent or tampon; **fluid spills into the field when the VVF is reached**; a cystoscopically placed catheter through the tract and ureteric stents help.
- **Indocyanine green (ICG) with near-infrared imaging** assesses vascularity of the bladder and vaginal edges — an advantage of minimally invasive repair.
- **Outcomes:** very high laparoscopic closure even largely without interposition; robotic extravesical repair (first with sigmoid epiploic appendages as interposition) gives similar closure. Concomitant **ureteric reimplantation or Burch colposuspension** is possible.
- **Transvesical versus extravesical: equal success** (cohort and systematic review); transvesical **shorter operative time**. Transvesical suits **low-lying fistulae** and avoids adhesiolysis; **extravesical suits supratrigonal VVF** and minimises new suture lines.
- **Single-port** laparoscopic and robotic repair is feasible; data too scant for conclusions.

TABLE 121.8 INSTRUMENTS FOR A ROBOTIC VVF REPAIR

Phase	Arm 1 (left hand)	Arm 2	Arm 3	Arm 4 (right hand)	Accessory port
Port	Robotic 8-mm trocars				5 mm if sutures pass through a robotic trocar; 8 or 12 mm if suture exchange expected
Entry and dissection	ProGrasp	Maryland bipolar or fenestrated bipolar	0° or 30° three-dimensional camera	Monopolar curved scissors	Grasper; suction-irrigator for dissection
Closure	ProGrasp	Needle driver	Camera	Needle driver (or SutureCut)	Grasper or shears

- **Sutures:** vagina (one or two layers) **2-0 or 3-0** barbed polyglyconate or unbarbed polyglactin 910; bladder (two layers) **3-0 and 2-0**.
- **Accessories:** vessel sealer for an omental J-flap; cystoscope with working channel for stenting and fistula catheterisation; vaginal stent or hand-made tampon (glove with laparotomy sponges on ring forceps).
- Arm order varies with training and handedness; the accessory port goes lateral to arm 1 or 4 or between arms.

TABLE 121.9 ROBOTIC O'CONOR APPROACH

Step	Sequence
1 Positioning	Dorsal lithotomy, arms optionally tucked → bimanual → prep abdomen, perineum, vagina (all accessible) → cystoscopy → tag the tract: right-angle or Kelly clamp from the vagina, ureteric catheter fed to it and pulled through (valuable for very small fistulae) → stent ureters if close → urethral catheter → vaginal stent or tampon (keeps pneumoperitoneum, aids dissection)
2 Ports and docking	4–6 ports (3–4 robotic, 1–2 accessory); Xi ports in a horizontal line, camera at or above umbilicus → open or Veress entry, high-flow low-pressure pneumoperitoneum (Palmer's point after prior surgery) → survey → left paramedian 8 mm (8–10 cm, one hand's breadth, from midline), far-left 8 mm, right paramedian 8 mm, far-right assistant → Trendelenburg $\leq 30^\circ$ → laparoscopic steps first (adhesiolysis, omental J-flap ; gastric mobilisation rarely needed) → dock preferably from the side to keep vaginal access
3 Dissection and excision	Start the vesicovaginal dissection as in sacrocolpopexy , backfill 150–300 mL + stent → incise peritoneum → blunt and sharp separation → bladder and vagina often fused, risk of new perforations → open the dome and extend the cystotomy down to the fistula (the O'Connor advantage) to find the plane → staying in the plane protects both blood supplies → remove fistula catheter; dissect 1–3 cm beyond the tract → excise orifice; freshen or excise edges but avoid excess vaginal excision (shortening, tension)
4 Vaginal closure	2-0 absorbable, usually one layer, transverse to avoid narrowing
5 Interposition	Secure graft before bladder closure (better view): omental J-flap, pericolic epiploic appendage or peritoneal flap, fixed beyond the vaginal closure without tension (multiple points or barbed running suture)
6 Bladder closure	One or two layers 2-0 or 3-0 → first running suture from the inferior end, transvesically "from inside" (orifices always in view) → second from the upper end extravesically to meet it → vertical closure, perpendicular to the transverse vaginal line
7 Integrity	Fill 150–300 mL → inspect → undock → remove ports under vision → close fascia of ports ≥ 10 mm
8 Postoperative	Urethral catheter ± stents (stents out at 4–6 weeks) → drainage ≥ 10–14 days , then cystogram → liberal anticholinergics or β_3 -agonists → same-day or next-day discharge; ambulate, hydrate, normal diet within 24 hours

Figure 121.8: Xi ports in a line; open Hasson entry for the camera at the umbilicus; 12-mm assistant port lateral to arm 4.

Concept — the O'Connor cystotomy is not an end in itself: opening the bladder down to the fistula is the surest way to find the fused vesicovaginal plane, and the transverse-vaginal, vertical-bladder closure keeps the two suture lines crossing rather than overlying.

INTERPOSING FLAPS AND GRAFTS

- **Rationale:** interposed tissue **reinforces the repair, fills dead space and imports vascular and lymphatic supply.**
- **Widely accepted for complex VVF:** radiation-induced, recurrent, ischaemic obstetric, or any questionable tissue.
- **Vaginal route:** Martius labial fibrofatty pad or peritoneal flap. **Abdominal route:** greater omentum, peritoneum, epiploic appendages.

Transvaginal Interposing Options

- **Martius flap (1928)** — for **mid-to-distal vaginal** fistulae. Labial fibrofatty tissue supplied by the **internal pudendal artery inferiorly** and the **external pudendal artery superiorly**, with lateral **obturator** branches. **Dual pedicles** allow harvest in either direction to reach trigone, bladder neck or urethra.

TABLE 121.10 THE MARTIUS FLAP PROCEDURE

Step	Sequence
1 Labial incision	Harvest after the first two VVF layers, before advancing the vaginal flap → mark and infiltrate local anaesthetic with adrenaline → vertical labial incision about 4–7 cm → borders: labiocrural fold laterally; bulbocavernosus and labium minus medially; Colles fascia over the urogenital diaphragm posteriorly
2 Fat pad dissection	Babcock or Allis on the yellow fibrofatty pad → relatively avascular areolar planes, blunt and sharp → stay lateral to bulbocavernosus and ischiocavernosus → free the deep surface down to the adductors
3 Creating the flap	For VVF, keep the superior supply (external pudendal labial vessels) and ligate the inferior end for reach to the anterior vagina (the reverse for rectovaginal fistula) → graft of 8–15 cm
4 Tunnelling	About 2 cm wide subepithelial tunnel from labium to repair → clamp passed from the vaginal side draws the pad through
5 Fixation	Over the layered closure with 3–4 interrupted 2-0 or 3-0 absorbable sutures , tension-free → vaginal epithelium advanced over all layers
6 Labial closure	Haemostasis → layered closure (no dead space) → monofilament or skin glue → Penrose drain 48 hours and pressure dressing

- **Outcomes:** high closure with Martius in non-irradiated and radiation-induced VVF. **RCT in simple VVF: no benefit** of adding Martius, and nearly a third had **donor-site pain or numbness.**
- **Trap:** Martius adds morbidity without benefit in **simple VVF** — its place is the complex fistula.
- **Peritoneal flap** — for **high (apical, post-hysterectomy) VVF** beyond Martius reach. Posterior dissection carried past the bladder to the **peritoneal fold of the pouch of Douglas** (glistening, translucent) → reflection freed off the posterior bladder without opening it (an inadvertent peritoneotomy is closed) → advanced over the repair and fixed with 2-0 distal to it on vaginal or perivesical fascia → vaginal epithelium advanced over it (Figure 121.9, Raz 2002).
- **High closure** with peritoneal flaps; flaps were used for VVF **≥2 cm, radiation-induced or recurrent.**

- **Full-thickness labial, Singapore and gracilis myocutaneous flaps** are reserved for refractory or complex VVF with **large soft-tissue defects**.

TRANSABDOMINAL INTERPOSING OPTIONS

- **Omental flap** — introduced by Kiricuta (with Goldstein) for **post-radiation VVF**. Supplied by **both gastroepiploic arteries** → mobilises deep into the pelvis on a robust pedicle; possesses **lymphatic properties** aiding healing and re-epithelialisation.
- **Evidence:** in a mixed benign and malignant series, repairs with omentum closed more often than those without. In **simple VVF**, omental interposition **did not change cure** (comparative study). Also used for large and obstetric VVF.
- **Peritoneal flap** (when omentum is scant or will not reach): peritoneum over the iliac fossa beside the bladder, or peritoneum and fibrofatty tissue directly behind the bladder inverted between bladder and vagina — **quicker than an omental J-flap**; open or minimally invasive.
- **Colonic epiploic appendages:** readily available, **no dissection needed**.
- **Rectus abdominis myofascial flap:** radiated and non-irradiated VVF, but morbid and demanding — **last resort**.
- **Other materials:** fleece-bound sealing material, dehydrated amniotic membrane, fibrin glue.

INTERPOSITION OPTIONS COMPARED

Flap	Route	Blood supply / feature	Best use
Martius	Vaginal	Internal pudendal (inferior), external pudendal (superior), obturator (lateral)	Mid-distal VVF, bladder neck, urethra; not the apex
Peritoneal (vaginal)	Vaginal	Pouch of Douglas reflection	Apical post-hysterectomy VVF
Gracilis, Singapore, labial	Vaginal / perineal	Myocutaneous or fasciocutaneous	Refractory VVF; large tissue loss
Omentum (J-flap)	Abdominal	Gastroepiploic arteries ; lymphatic	Radiation, recurrent, large VVF
Peritoneal (abdominal)	Abdominal	Iliac fossa or retrovesical	Scant or short omentum; quick
Epiploic appendages	Abdominal	Sigmoid	Readily available, no dissection
Rectus abdominis	Abdominal	Myofascial	Last resort

Concept — *interposition earns its place where the host tissue cannot heal on its own — radiation, recurrence, ischaemia — and randomised data show it adds nothing but morbidity in the simple fistula; the flap chosen is simply the best-vascularised tissue that reaches the repair.*

URINARY SYMPTOMS AFTER VESICOVAGINAL FISTULA REPAIR

- **Few studies look beyond closure.** Preoperatively, **concurrent SUI and detrusor overactivity (DO)** are both **common** in fistula patients; after repair, persistent symptoms were mainly **urgency and UUI**, occasionally SUI, despite anatomical cure.
- **De novo urinary symptoms** (UUI and SUI) occur in a minority. **UUI was more frequent after open than transvaginal repair**, attributed to the **bivalving and long suture lines of the O'Connor technique**; International Consultation on Incontinence Questionnaire – Short Form (ICIQ-SF) scores did not differ.

- Other comparisons: no difference by route in urodynamic SUI, OAB or UTI; validated questionnaires show **symptoms improve after both routes**; no difference between open and laparoscopic repair in UTI, SUI or spasms.
- **Postoperative retention** is uncommon; predicted by **vertical (versus horizontal) bladder closure**; prior repair status was also associated.
- **Residual incontinence after obstetric repair** is predicted by **Goh types 3 and 4**, more than one repair, a positive cough stress test and a positive pad test.
- **Residual incontinence drives QoL and mental health**: all patients without residual leakage reported improved QoL against about half of those with it; **every patient with a depressive disorder, and all with suicidal ideation, still leaked**.
- **International Continence Society (ICS)**: recommends **pelvic floor physical therapy (PFPT)** for all obstetric fistula patients and for persistent symptoms after repair; access is limited worldwide, but risk is low.

Concept — *anatomical closure is not cure: the patient's experience is set by residual incontinence, which is predicted by how near the fistula lay to the continence mechanism and how much repair the bladder has needed.*

SEXUAL FUNCTION AFTER VESICOVAGINAL FISTULA REPAIR

- **Little is known**. On the **Female Sexual Function Index (FSFI)** (19-item validated), overall sexual function **improved after both routes**, but **pain, arousal and orgasm domains did not** — attributed to vaginal pain and dyspareunia.
- **After transvaginal repair in Malawi**, a minority reported intercourse problems (half from pain, half from persistent incontinence). The Female Sexual Distress Scale–Revised (FSDS-R) improved overall, but some stayed in the dysfunction range; **reduced vaginal calibre and fistula size ≥ 3 cm** predicted dysfunction.
- **ICS Physiotherapy Committee**: pelvic floor relaxation for dyspareunia, vaginismus and hypertonicity after any fistula repair; cost and access limit it.

URINARY DIVERSIONS FOR THE TREATMENT OF VESICOVAGINAL FISTULA

- **When repair is unfeasible or unwise**: multiple failed attempts, coexisting abdominopelvic disease, **severe radiation damage**, massive soft-tissue loss, comorbidity.
- **Fit for surgery but not for repair** → **conduit or continent reservoir**.
- **Unfit for any surgery** → **percutaneous ureteric occlusion with permanent nephrostomy**.
- **Resource-poor settings** (no catheters or stoma appliances) → **ureterosigmoidostomy** as a last resort, accepting its metabolic and neoplastic risks.

URETEROVAGINAL FISTULA

- **Ureterovaginal fistula (UVF)** — ureter-to-vagina connection, **most commonly from unrecognised iatrogenic injury to the distal third of the ureter** during pelvic surgery.
- Rarely from obstetric or infertility procedures, malignancy, radiation, endometriosis, inflammation or infection, retained foreign bodies.
- **Typically follows a mundane benign hysterectomy with no untoward intraoperative event**.

Aetiology

- **About three-quarters follow unrecognised injury at abdominal hysterectomy**, owing to the intimate relation of the **uterine artery and the ureter**.

- **Ureteric injury at hysterectomy: 0.2–2.4%.** Registries disagree on route: one found GU injury commonest after **abdominal** hysterectomy; another found ureteric injury commonest after **laparoscopic** and rarest after **vaginal**; a population study found no difference in delayed injury by route. Radical hysterectomy is assumed higher risk, though published incidences are similar.
- **Mechanisms:** laceration, partial or complete transection, avulsion, crush, thermal injury, suture ligation, **ischaemia from devascularisation**.
- **Diagnosis is delayed in 18–88%.** Routine cystoscopy series detect several times more ureteric injuries than series without it — cystoscopy uncovers injuries that would otherwise go unrecognised.
- **Delayed detection raises ureteric fistula risk about fourfold; combined bladder and ureteric injury** raises fistula risk further, dramatically so when detection is delayed.
- **Up to 25% of VVF have a coexisting UVF.**

Prevention

- **The optimal goal is prevention**, founded on pelvic anatomy.
- **Sites of injury:** (1) pelvic brim, medial to the infundibulopelvic ligament; (2) where it passes under the uterine artery in the cardinal ligament; (3) lateral to the uterosacral ligaments; (4) over the anterolateral vaginal fornix just proximal to the ureterovesical junction.
- **Most injuries are in the distal 6 cm, slightly favouring the left** — and these distal injuries cause UVF.
- **Risk factors:** Black race (self-reported), prior abdominal surgery, **endometriosis**, route of surgery, **concomitant apical suspension**, longer operation, lower body mass index (BMI), **blood loss ≥800 mL**, uterine size. After radical hysterectomy: prolonged surgery, **diabetes**, American Society of Anesthesiologists (ASA) 3–4, postoperative infection.
- **Preoperative IVP does not reduce injury** and is not cost-effective (hundreds of IVPs per injury prevented).
- **Prophylactic ureteric stents — controversial.** For: haptic feedback, rapid recognition. Against: iatrogenic injury, UTI, time. An RCT in major gynaecological surgery showed **no reduction**; a later meta-analysis showed a small reduction with a **number needed to treat of 224**; cost-effective only if injury risk **>3.2%**. In **caesarean hysterectomy for placenta accreta spectrum**, stents reduced bladder and ureteric injury → **selective, not routine, use.**

Clinical Presentation and Diagnosis

- **Consequences of unrecognised injury:** flank pain, anuria, ileus, peritonitis, sepsis, stricture, fistula, loss of renal function.
- **Commonest presentation: continuous leakage, usually within 4 weeks of surgery**, ± fever and flank pain; in retrospect a difficult recovery (ileus, fever, slow progress).
- **Key contrast with VVF: normal voiding plus constant wetness**, because the contralateral ureter fills the bladder. **Exceptions: bilateral injury or concomitant VVF.**
- **Confirm:** pooled vaginal urine (creatinine); **double dye test** (VVF, UVF or both); sometimes a visible defect.
- **Upper tract:** CTU, IVP or retrograde pyelogram. Urography commonly shows **hydronephrosis, hydroureter or a non-functioning kidney — none pathognomonic.** **Ultrasound alone is not definitive** (no excretory phase), although absence of grade 3 hydronephrosis has a high negative predictive value in obstetric fistula.
- **Retrograde pyelogram:** abrupt termination of contrast, or **ureteric continuity with a fistula → attempt stenting.**
- **Exclude coexisting VVF:** cystogram with oblique and lateral views; **CTU and MRI show both and are superior** to plain radiography and ultrasound.

VVF VERSUS UVF — THE GOVERNING CONTRAST

	VVF	UVF
Voiding	Reduced or absent desire (bladder never fills)	Normal voiding plus constant leak
Onset after hysterectomy	Days (cystotomy) or 7–10 days (thermal/crush)	Usually within 4 weeks, often with fever and flank pain
Upper tract	Usually normal (image to exclude UVF)	Hydronephrosis or obstruction common
First line	Catheter drainage (selected), then repair	Ureteric stent , then reimplantation

Management

- **Goals: dryness and preservation of renal function.** Management has moved **from nephrectomy** (mid-20th century, about half of cases) → **reimplantation** → **percutaneous nephrostomy** → **endourology**.
- **Ureteric stenting is first line, as soon as possible** (retrograde, or antegrade via interventional radiology). **Stenting within 2 weeks of detection succeeds far more often than stenting at 2–6 weeks**; in candidates able to be stented, most resolve with stent alone (stents typically left in for weeks to a few months); stricture may follow.
- **If stenting fails or is impossible** → **surgery** (Figure 121.10). **Timing has no consensus** — early surgery risks oedema, inflammation and necrosis; delay prolongs distress. Early reimplantation took less time and blood loss with equal success; a systematic review found **equal success before and after 6 weeks**.
- **Ureteroneocystostomy is usually feasible** (distal third): dissect the ureter as distally as possible for length → mobilise the bladder → **psaos hitch** to reduce tension → **Boari flap** for longer defects → preserve periureteric adventitia, **especially laterally where the blood supply enters**. Alternatives for length: **downward nephropexy, ileal ureter**.
- **Success ≥90% (pooled near 100%)**. Laparoscopic and robotic repair match open; **vaginal reimplantation** (with psaos hitch or Boari flap) has been reported; spontaneous closure also occurs.
- **Complications:** hydronephrosis, infection, **stricture (ischaemic)**.
- **Follow-up (no consensus): ultrasound 1–2 weeks after stent removal**; some add a **renal scan at 6 weeks**; stricture usually appears within a year but **up to 30 months** → follow beyond this.

FIGURE 121.10 — ALGORITHM FOR URETEROVAGINAL FISTULA

Stage	Action
Suspect UVF	Exclude VVF (cystoscopy ± VCUG, double dye test) → confirm (IVP ± retrograde pyelography [RPG] / CT)
Attempt stent	Successful → remove at 4–6 weeks → repeat imaging → resolved, or persistent → surgery
Stent fails	Surgical repair (ureteroneocystostomy)

Concept — *UVF is a distal ureteric injury that drains past a normally filling bladder, so it presents as wetness with normal voiding; because the ureter heals over a stent if reached early, the clock from detection to stent largely decides whether the patient needs an operation.*

VESICOUTERINE FISTULA**Aetiology**

- **Vesicouterine fistula (VUF)** — bladder–uterus connection; historically **about 5–7% of GU fistulae**, but rising sharply where caesarean rates rise (it made up a large share of fistulae in a recent Rwandan series).

- **The vast majority follow a low transverse caesarean delivery**; operative vaginal delivery also. Other causes: hysterectomy, intrauterine devices, dilatation and curettage, induced abortion, manual removal of placenta, uterine rupture, traumatic catheterisation, brachytherapy, uterine embolisation.
- **Mechanism: apposition of an unrepaired uterine incision to an occult bladder injury** (for example, at hysterotomy closure), or **incorporation of the bladder in the hysterotomy closure**.

Clinical Presentation

- **Usual site:** uterine corpus just above the internal os ↔ posterior bladder.
- **Continence is usually preserved** because the **cervix is competent**.
- **Youssef syndrome: cyclical haematuria (menouria), amenorrhoea, first-trimester miscarriage and absence of incontinence.**
- **Incontinence occurs** if the tract lies **below the internal os** or the cervix is incompetent (common in some series).
- **Timing:** direct injury presents early; obstructed labour or irradiation presents later (necrosis).
- **Differential of menouria: bladder endometriosis** (a fifth to a quarter develop menouria) and **congenital anomalies**.

Diagnosis

- **Dye test:** blue saline from the **cervical os** after bladder backfill.
- **Saline-infusion sonohysterography and hysterosalpingography** may show the uterine defect and contrast entering the bladder — operator dependent. **Retrograde cystogram** shows extravasation into the uterine cavity. **MRI** confirms.
- **Excretory urogram is required** to exclude concomitant upper tract fistula — **MR urography** can serve both purposes.
- **Cystoscopy** (indicated for haematuria) shows the defect, usually **supratrigonal posterior**, and its relation to the ureters; the scope can often enter the uterine cavity — **seeing both tubal ostia is highly suggestive**.

Management

- **Usually surgical — spontaneous closure is rare** (about 5%, possibly hormone related). Fulguration plus prolonged catheterisation has closed an isolated case.
- **Timing is controversial:** early for psychosocial reasons, or **≥3 months** for inflammation; **after caesarean, wait 3–4 months for full uterine involution**; earlier repair works but is harder.
- **Approach: abdominal, increasingly minimally invasive.** Transperitoneal access allows **complete separation of bladder and uterus for a tension-free bladder repair**.
- **Fertility wishes decide the operation** (table below); uterus-sparing repair follows VVF technique with omentum or peritoneum between bladder and uterus; with hysterectomy, tract excision is optional.
- **Transvaginal repair** (commoner before 1947): dissection as for vaginal hysterectomy → vesicocervical then vesicouterine spaces → tract → bladder mobilised for two-layer tension-free closure → uterine defect closed **perpendicular** to the bladder → colpotomy closed; or vaginal hysterectomy. **Martius flap is generally not feasible** (too far from the introitus).
- **After uterus-sparing repair** pregnancy and term birth are achievable; the ideal contraceptive interval is unknown, caesarean delivery rests on extrapolated rupture risk, and **recurrence reaches about 20%** over long follow-up.

VUF — UTERUS-SPARING VERSUS HYSTERECTOMY

	Uterus-sparing	With hysterectomy
Patient	Desires fertility	Childbearing complete
Technique	O'Connor or extravesical; interposition	Hysterectomy then two-layer bladder closure
Afterwards	Contraception until healed; caesarean delivery; recurrence risk	—

Concept — *because the competent cervix holds urine in the uterus, VUF presents not as incontinence but as blood in the urine at menstruation — Youssef syndrome — and continence appears only when the tract lies below the internal os.*

URETHROVAGINAL FISTULA**Aetiology**

- **Developing world:** overwhelmingly **obstructed labour**, often with a VVF; urethral and continence-mechanism damage increases complexity and failure (as Goh and Waaldijk encode). Nearly half of GU fistulae in one Ethiopian series were urethral or juxtaurethral. Others: radiation, pelvic trauma, traumatic catheterisation, malignancy.
- **Developed world:** **iatrogenic** — **diverticulectomy**, anti-incontinence surgery, anterior repair with or without mesh.
 - **Urethrovaginal fistula complicates about 1–8% of diverticulectomies; diverticulectomy accounts for 20–25% of urethrovaginal fistulae.**
 - **Slings (autologous and synthetic) are the commoner anti-incontinence cause, but periurethral bulking** has also been implicated.
 - Also pessaries, gender-affirming surgery, feminising genitoplasty and cloacal repair in children (early or late), and **pressure necrosis from an indwelling catheter** (bladder neck and urethral destruction).

Clinical Presentation

- **Symptoms depend on position relative to the sphincter and on size: distal to the sphincter → largely asymptomatic; proximal near the bladder neck → near-constant leakage or SUI.**
- Split or splayed stream, **vaginal voiding**, post-void dribble on standing; local pain, dyspareunia.
- **After mesh:** bleeding, partner awareness during intercourse (**hispareunia**); fistula may be found incidentally with mesh exposure.
- Also perineal skin irritation, recurrent UTI, vaginal candidiasis.

Diagnosis

- **Speculum examination usually suffices** (distal site); assess coexisting prolapse, SUI, atrophy for simultaneous treatment. Prominent rugae may hide it → **methylene blue dye test**.
- **VCUG (lateral view)** shows vaginal contrast; with VUDS it characterises associated incontinence.
- **Cystourethroscopy is essential** — size, location, **exposed synthetic material**. **Flexible scopes are better for the urethra** because outflow is at the lens; rigid scopes eject fluid 1–2 cm proximal to the lens and distend the lumen poorly. Also evaluates bladder, bladder neck, orifices and other fistulae.
- **About 20% coexist with a separate VVF** — search for it. **Evaluate the upper tract.**

Management

- **Timing debated** — early if tissue allows, or delay until inflammation settles.
- **Among the most challenging reconstructions:** the difficulty is less the location than the **scarcity of viable tissue for flaps**, especially after trauma or radiation. Historically, urethral destruction was the reason the few irreparable fistulae in a large series needed ureterocolonic diversion to achieve continence.
- **Standard: transvaginal repair with rotational vaginal flaps** (Table 121.11). Primary urethral repair is occasionally possible, but **most repairs add tissue — Martius flap most often**, occasionally omentum or gracilis; an **autologous rectus fascia patch** is an alternative.
- **Concomitant autologous (pubovaginal) sling** for preoperative SUI or high SUI risk (bladder neck or sphincter involvement) — **synthetic material is avoided because it may erode the poorly vascularised urethra**.
- **Near-total urethral loss: tubularised anterior bladder flap** as a neourethra; a posterolateral bladder tube has also been used.
- **With prior mesh: remove all mesh visible in the repair field.**
- **Position:** usually dorsal lithotomy; **prone** can improve urethral and anterior wall exposure.
- **Outcomes:** anatomical closure **90–95% with vital-tissue adjuncts**, but **continence only 50–80%**; postoperative SUI can reach half; some need further SUI surgery even after a concomitant sling.
- **Strategy for continence: simultaneous** sphincter repair with Martius flap and pubovaginal sling, **or staged** — close the fistula first, treat residual symptoms later.
- **Other complication:** urethral stricture (dilatation or reoperation).

TABLE 121.11 URETHROVAGINAL FISTULA REPAIR USING A TRANSVAGINAL ROTATIONAL FLAP

Step	Sequence
1 Positioning and exposure	Dorsal lithotomy → bimanual → prep → weighted speculum → cystourethroscopy; cannulate tract (guidewire / open-ended catheter) → optional suprapubic via Lowsley tractor → urethral catheter → episiotomy if narrow → ring retractor
2 Incisions	Infiltrate saline or dilute adrenaline → circumscribe tract with 1–2 mm cuff; minimal excision so the tract is not enlarged → J- or U-incision with the curve just proximal to the tract → raise vaginal flap
3 Flaps	Expose periurethral tissue to the bladder neck or beyond; free laterally and distally → optional second periurethral flap (stay superficial to mucosa) or simple imbrication → if a sling is planned, open the retropubic space lateral to the bladder neck (before or after the repair)
4 Tract closure	Replace catheter with a urethral sound (protects catheter and avoids catching the dorsal urethra) → 4-0 absorbable over the tract, transverse (against stenosis) or longitudinal → second layer perpendicular , 4-0 or 3-0 → large defect: buccal mucosa graft or vaginal flap onlay → test with saline ± methylene blue via angiocatheter → 16–18 Fr catheter
5 Vaginal flap	Adjunct layer or sling first (Martius, abdominal or fascia lata fascia, gracilis or rectus muscle, myocutaneous flap, full-thickness labial graft) → advance vaginal flap beyond the closure, 2-0
6 Postoperative	Pack 24 hours only if bleeding → with suprapubic, urethral catheter out at 2–3 days; suprapubic ≥14 days (or both catheters throughout) → lateral VCUG before suprapubic removal → nothing per vagina for 6 weeks

Figures 121.13–121.15: fistula at the bladder neck after a synthetic mid-urethral sling, with calcified intraurethral sling; fistula after endoscopic bladder neck suspension; repair (Leach and Kobashi, 2001) through an **inverted-U incision based at the proximal fistula margin**, with the **tract edges not excised**.

Concept — *in urethrovaginal fistula closure is the easy outcome and continence the hard one: the lesion sits in or beside the sphincter, so success depends on vital tissue for the repair and autologous — never synthetic — support for the urethra.*

UROENTERIC FISTULAE

Rectourethral Fistula

Aetiopathogenesis and Epidemiology

- **Rectourethral fistula (RUF)** — lower urinary tract to distal rectum; rare; **congenital** (a subgroup of anorectal malformation) or **acquired** (trauma, neoplasm, infection, inflammation).
- **Most acquired RUFs are iatrogenic** — radical prostatectomy (RP), cryosurgery, external beam radiotherapy or brachytherapy. Historically TURP (transurethral resection of the prostate) and open simple prostatectomy; **today mostly prostate cancer (CaP) treatment**.
- **Burden:** raised mortality and morbidity, poor QoL, long stay, high cost.
- **Symptoms:** **faecaluria and pneumaturia** (most typical); other gastrointestinal disturbance common; clinically **recurrent cystitis and dysuria**.
- **Radiation-induced RUF:** commonest after **combined external beam and brachytherapy**, then brachytherapy, then pelvic irradiation (prostate or anal cancer).
- **Non-radiation RUF:** mainly surgery or instrumentation (**most after prostatectomy**), then trauma (penetrating, pelvic fracture, road traffic), then inflammatory bowel disease.
- **Rectal injury during RP by approach:** highest with **retropubic and laparoscopic**, lower with perineal, **lowest with robot-assisted**; a United States population study likewise ranked **open > laparoscopic > robotic**.
- **Other energies:** primary whole-gland cryotherapy — low and falling RUF rate with better selection; primary high-intensity focused ultrasound (HIFU) — low; **salvage HIFU after radiation — up to about 9%**, much lower when post-radiotherapy selection parameters are used.
- **Females:** RUF is **virtually impossible because the vagina lies between urinary tract and rectum**.
- **The greatest challenge is the recurrent fistula.**

MARTINI CLASSIFICATION OF RUF AFTER PROSTATE CANCER TREATMENT

Axis	Category	Definition
Stage (size, on CT urography or cystography)	I	≤1.5 cm
	II	>1.5 cm
	III	Any size with urethral sphincter damage
Grade (state of surrounding tissue)	0	Inadvertent rectal injury at surgery, no prior non-surgical treatment
	1	Prior non-surgical treatment — radiation, chemoradiation, brachytherapy, HIFU or other energy
	2	Salvage RP or salvage ablation

- **Pattern:** stage describes the **hole**; grade describes the **host tissue** — and tissue, not size, chiefly governs repair.

Diagnosis

- **Digital rectal examination** may feel the fistula on the anterior rectal wall.
- **Cystoscopy and sigmoidoscopy** show the tract, allow **biopsy to exclude recurrent malignancy** and assess tissue vitality.
- **VCUG or retrograde urethrography (RUG) is usually conclusive** — size and location for planning. **Image the upper tract** to exclude ureteric injury. **Pelvic MRI** for complex cases.
- **Assess continence and sphincter function** (elderly, post-RP): repairing the RUF alone often fails to restore continence in severe SUI; an **artificial urinary sphincter (AUS)** is possible later but with **higher failure and erosion**. UDS or VUDS with **rectal packing** help when leakage limits filling.

Treatment

Conservative therapy.

- **If no sepsis and no faecaluria:** low-residue diet with urinary and/or faecal diversion (urethral or suprapubic catheter, nephrostomy, ileostomy, colostomy).
- **A simple fistula closes in about 12 weeks; persistence beyond this → surgery. Complex and irradiated RUF rarely close** → conservative care inappropriate.
- **Post-RP series:** small fistulae without faecaluria closed with prolonged catheterisation (weeks to months); early post-laparoscopic RP fistulae healed with catheter and intravenous nutrition, some needing temporary colostomy.
- **Bottom line:** predictors are unknown; conservative care is a **reasonable first step in most**, while inflammation settles (**≥12 weeks**, debatable) and preparatory steps (colostomy) are taken.

Surgical treatment.

- **No universally accepted operation** — the number of techniques reflects rarity and difficulty. Decision factors: surgeon preference, fistula location, prior failed repair, **radiation**.
- **A diverting colostomy usually precedes repair, whatever the technique.**

RUF REPAIR APPROACHES COMPARED

Approach	Essentials	Strengths	Weaknesses
Perineal (Young 1917; large series by Goodwin 1958)	Dissect tract; close bladder/urethral and rectal defects separately; gracilis flap between urethra and rectum (or buccal graft, dartos flap)	High closure; most stomas reversed; high satisfaction	Significant urinary incontinence common
Transsphincteric (York-Mason)	Prone jackknife; coccyx-to-anal-verge incision; posterior sphincter divided in layers; rectal mucosa opened; tract excised; three layers (urothelium, rectal muscle, rectal mucosa) closed separately	Avoids lateral pelvic and pararectal spaces → preserves sexual function, continence and rectal innervation; high success, sometimes after repeat procedures	Poor bulbar/membranous exposure; no healthy flap except rectal advancement; dehiscence and infection; faecal incontinence up to 20%
Open abdominal	Transvesical, extravesical or mixed O'Connor, as for VVF	Omental or peritoneal flap; suits large, irradiated RUF or cases needing extensive resection or permanent diversion; good closure	Morbidity, slow recovery, poor deep-pelvic view, hard after prior surgery

Approach	Essentials	Strengths	Weaknesses
Laparoscopic / robotic	Transperitoneal; open pouch of Douglas; dissect to orifice; cystotomy if needed (O'Connor-like); close rectum in two transverse layers; peritoneal or omental flap ; close cystotomy; test bladder and rectum	Deep-pelvic vision ; combinable with perineal approach (gracilis; side-docking robot plus perineal surgeon) — avoids sphincter division and pubectomy	Limited outcome data (small series without recurrence)

- **Abdominal technique trade-offs (as for VVF):** transvesical — hard to bring interposition and to divide rectal and urethral walls widely; extravesical — ureteric injury risk, difficult rectourethral dissection and urethral closure; O'Connor-like — larger opening of the urinary tract and possible voiding dysfunction.
- **Irradiated or ablated RUF** are mostly repaired transabdominally, because of severe radiation damage, large size or need for extensive resection and permanent diversion; **cystoprostatectomy with ileal conduit** is the fallback when repair is impracticable.

Concept — *RUF repair is governed by the host tissue: a fresh post-prostatectomy injury may close with diversion or a perineal or transsphincteric repair, whereas an irradiated fistula needs imported vascularised tissue from the abdomen or thigh — or diversion.*

Vesicoenteric Fistula

Aetiopathogenesis and Epidemiology

- **Vesicoenteric fistula (VEF)** — bowel–bladder communication; infrequent but highly morbid.
- **Types:** colovesical (**commonest**), rectovesical, ileovesical and **appendicovesical (<5%)**.
- **Simple:** small, solitary, non-irradiated. **Complex:** larger, multiple tracts, irradiated tissue, often with **pelvic abscess or colonic obstruction**.
- **Sixth and seventh decades; male : female 3 : 1** — **the uterus shields the bladder from the sigmoid; prior hysterectomy is frequent in affected females.**
- **Causes:** **diverticular disease (commonest)**, then **Crohn disease**; malignancy (mainly colorectal, also bladder and prostate) a small share; appendicitis, trauma, radiation, iatrogenic.
- **Frequencies:** fistulae in **4–23%** of those admitted with diverticular disease; colovesical fistula (CVF) in **2–23%** of diverticular disease; VEF in **2–8% of Crohn disease**.
- **Urinary symptoms dominate:** **pneumaturia (50–70%) and faecaluria (up to 51%)**; also frequency, urgency, suprapubic pain, **recurrent UTI** (*Escherichia coli*, coliforms, mixed growth, enterococci), haematuria. **Over 75%** have pathognomonic features.
- **Gouverneur syndrome** — the distinctive picture: **suprapubic pain, frequency, dysuria and tenesmus.**

Diagnosis

- **No accepted gold standard.**
- **Cystoscopy:** usual but findings (erythema, oedema) nonspecific; detection varies widely (35–90%).
- **Colonoscopy:** poor at showing the fistula, but **10–15% of CVFs have an underlying neoplasm** → defines the bowel disease.
- **Poppy seed test:** ingest poppy seeds → inspect urine voided over **48 hours**; seeds in urine strongly indicate a uroenteric fistula. **Highest detection of any test in direct comparisons**, cheap and simple — but gives **no location or type**.
- **Ultrasound with compression:** echogenic "**beak sign**" joining peristaltic bowel to bladder; reasonable sensitivity in Crohn internal fistula; operator dependent, rarely used.

- **CT is the preferred imaging method** — best for the extravesical disease. Signs: **air in the bladder**, focal wall thickening, extravesical mass, bladder adherent to bowel, **oral or rectal contrast in the bladder** (Figure 121.17).
- **MRI**: excellent soft-tissue contrast and multiplanar anatomy; used in inflammatory bowel disease and CVF.
- **Barium enema and cystography**: low detection (about 30%).
- **Technetium-99m diethylenetriaminepentaacetic acid (Tc-99m DTPA)** scintigraphy shows radioactive urine entering bowel and grades flow; not yet routine.

VEF DIAGNOSTIC TESTS COMPARED

Test	Role	Limitation
Poppy seed test	Best at confirming presence	No location or type
CT	Preferred imaging; shows cause and extravesical disease	—
Cystoscopy	Routine; nonspecific signs	Variable detection
Colonoscopy	Excludes neoplasm	Poor fistula detection
MRI	Anatomy, inflammatory bowel disease	—
Barium enema, cystography	—	Low yield
Ultrasound, Tc-99m DTPA	Beak sign; functional flow	Operator dependent; not routine

Treatment

Conservative treatment.

- **Candidates**: minimal symptoms, non-malignant cause, **especially Crohn disease** — bowel rest, total parenteral nutrition, antibiotics, steroids, immunomodulators, urethral catheter.
- Untreated CVFs in one series caused symptoms but **no urosepsis, no renal decline and no difference in disease-specific mortality**.
- Nevertheless, given morbidity, **surgery is usually offered to fit, informed patients; spontaneous closure is very rare (<2%)**.

Surgical treatment.

- **Choice depends on pathology, location, fitness and surgeon.**
- **Single-stage**: resection with primary anastomosis, no stoma. **Multistage**: resection plus colostomy and/or **Hartmann procedure** (two-stage), then stoma closure (three-stage) — for **gross faecal contamination, large pelvic abscess, advanced malignancy or radiation change**.
- **Resect the diseased colonic segment together with the fistula** — essential to prevent recurrence. Simple suture of bladder and colon defects with omental interposition has also been used.
- **Bladder side**: excision versus oversewing is unimportant; **small defects may be left to heal**; in benign disease **bowel resection alone usually suffices**, and no repair is needed if no bladder defect is seen or felt. **Malignancy → partial cystectomy with adequate margin**.
- **Laparoscopy**, once contraindicated in diverticular fistula (inflammation, long operations, conversion), is now safe in selected cases: the commonest operation is **one-stage resection with primary anastomosis** (±

loop ileostomy); minimally invasive attempts convert infrequently, with low severe complication, mortality and recurrence rates; laparoscopic CVF series report no conversions and no long-term recurrence.

Concept — VEF is a bowel disease presenting in the bladder: diagnosis rests on proving the connection (poppy seed) and defining the bowel pathology (CT, colonoscopy), and cure rests on removing the diseased bowel — the bladder usually heals itself.

Renocolic Fistula

- **Uncommon; historically more often from renal inflammatory or malignant disease than from colonic disease.** Divided into **traumatic and spontaneous**; spontaneous is commoner and usually renal in origin.
- **Xanthogranulomatous pyelonephritis (XGP)** and other renal infections extend into the pararenal space and involve bowel. Others: external or surgical trauma, renal or colonic carcinoma, diverticular disease.
- **Prerequisites: chronic renal disease with atrophy of perinephric fat.**
- **Colonic perforation at percutaneous nephrolithotomy (PCNL)** is rare (0.2–0.3%) but dangerous; fibrin sealants have closed a few persistent nephrocutaneous fistulae. **Except small instrumentation fistulae, treatment is surgical**, bowel resection ± anastomosis added as local conditions dictate (table).

RENOCOLIC FISTULA — IATROGENIC VERSUS SPONTANEOUS

	Small iatrogenic (PCNL)	Spontaneous (XGP, chronic infection)
Kidney	Usually functioning	Chronically diseased, often non-functioning
Management	Conservative first: antibiotics, nephrostomy or stent, nil by mouth	Nephrectomy ± bowel resection

UROCUTANEOUS FISTULA

Vesicocutaneous Fistula

Aetiopathogenesis and Epidemiology

- **Vesicocutaneous fistula (VCF) causes:** infection, neoplasm, bladder pathology (stones, diverticula, trauma), radiotherapy, and surgery — open prostatectomy, total hip arthroplasty, radical hysterectomy, **opening of an ileocystoplasty** (for example, for stones), one case after transobturator sling.
- **Commonest in routine practice: persistence after long-standing suprapubic cystostomy.**
- **Persistence factors:** well-epithelialised tract, infection, foreign body, neoplasm.
- Suprapubic fistula follows about 1% of open prostatectomies for benign prostatic hyperplasia (BPH). Tracts run mostly to the **thigh or anterior abdominal wall**.
- **Presentation:** abscess or cellulitis, dysuria, fever, UTI, suprapubic pain, watery wound discharge.
- **Exclude bladder outlet obstruction (BOO)** — raised intravesical pressure drives formation and persistence.

Diagnosis

- **Confirm with** cystography, retrograde fistulography, CT cystography, MRI (complex tracts).
- **Cystoscopy** shows irrigant leaking through the fistula, sometimes an orifice, and — crucially — **distal obstruction requiring treatment or bypass**. Methylene blue backfill or intravenous indigo carmine can help.

Treatment

- **Drain abscesses and give antibiotics.**

- **Most close with urinary diversion plus wound care**; bilateral nephrostomies occasionally for temporary diversion; **negative-pressure dressing** aids skin healing.
- **Non-healing (especially radiation)**: fistulectomy or **partial cystectomy**.
- **Relieve BOO** to lower bladder pressure.

Urethroperineal Fistula

Aetiopathogenesis and Epidemiology

- **Mostly after hypospadias repair**; less often congenital.
- **Iatrogenic urethrocuteaneous fistula (UCF)** is relatively common in developing countries because of **circumcision by unauthorised practitioners**.
- **Other causes**: pelvic fracture, infection, urethral or penile surgery (urethroplasty, Fournier debridement, priapism shunt, penile fracture repair), neoplasm, radiotherapy, gender-affirming surgery, distal obstruction (impacted stone).
- **UCF is the commonest complication of hypospadias surgery (up to 20%)**. Contributors: **distal obstruction, ischaemia, technical error, poor tissue handling, overlapping suture lines, infection**.
- **Recurrence after UCF repair about 25%**; some need up to 15 attempts; presentation can be **as late as 13 years**.
- **Congenital UCF** may accompany anorectal or genital malformations (ventral curvature, hypospadias) → multidisciplinary care.
- **Neurogenic self-catheterisation**: false passages in bulbar or membranous urethra plus pressure ulcers.

Diagnosis

- **Perineal dribbling** with a visible cutaneous orifice; sometimes recurrent UTI and LUTS.
- **RUG or fistulography** diagnose; **cystoscopy** shows the fistula and any stricture; **MRI** shows a fluid-filled tract from urethra to perineal skin.

Treatment

- **Goal: durable, recurrence-free and stricture-free urethra. The first repair matters most** — redo surgery faces distortion, scar and short urethral stumps.
- **Principles: reconstruct the urethra, add a waterproof layer, ensure skin cover** — often from several flaps.
- **Decision factors**: early or late, single or multiple, acute or chronic. **Early acute fistulae → conservative. Mature → surgery.**
- **Size rule: single fistula <1 cm → local tissue; ≥1 cm → regional skin and fascial flaps** with good blood supply, especially after repeated surgery.
- **After hypospadias repair wait ≥6 months** for inflammation to resolve.
- **Techniques for single post-hypospadias UCF**: tubularised incised plate (TIP) urethroplasty, Mathieu urethroplasty, double ligation, simple closure, trap-door — recurrence differed modestly; **choose by diameter, location and glans dehiscence**.
- **Large penoscrotal UCF (1–2 cm)**: three-layer repair with a **scrotal flap** — no recurrence in the reported series.
- **Covering layer**: recurrence after repair was lower with a **tunica vaginalis flap than with dartos fascia**. Scrotal flaps are rich in blood supply and easily advanced.
- **Extensive tissue loss (after Fournier debridement) → gracilis myocutaneous flap**.
- **Neurogenic patients with pressure ulcers → ileal conduit diversion** (± pressure-ulcer surgery) — excellent outcomes; may be the option of choice.

URETHROCUTANEOUS FISTULA — CHOOSING THE REPAIR

Situation	Choice
Early, acute	Conservative
Mature, single, <1 cm	Local tissue closure
≥1 cm or multiple redo	Regional skin and fascial flaps (scrotal)
Cover layer	Tunica vaginalis preferable to dartos
Massive loss (Fournier)	Gracilis myocutaneous flap
Neurogenic with pressure ulcer	Ileal conduit

Nephrocutaneous Fistula**Aetiopathogenesis and Epidemiology**

- **Renal fistulae to adjacent organs are not uncommon; kidney-to-skin is rare**, and rarer still without prior surgery.
- **Causes:** prior renal surgery, **stones (usually large or staghorn)**, trauma, tumours, chronic UTI including XGP. **Tuberculosis causes ≥10%** — consider it and its risk factors.
- **Drainage routes of least resistance: lumbar triangle of Petit and lumbar quadrilateral of Grynfeldt.**
- **The kidney is poorly or non-functioning in the vast majority** of nephrocutaneous fistulae (NCF).
- **Slight female predominance; left side commoner.**
- **Transplant:** calyceal-cutaneous fistula in about 3% of allografts, **12% with multiple renal arteries** — from segmental infarction.
- **Symptoms:** flank discharge (majority), loin pain (about one-third); flank sinus, occasionally a lumbar mass.

Diagnosis

- **Contrast-enhanced abdominal CT is preferred** — diagnosis, cause, renal function, planning. **Fistulography and retrograde pyelography** assess the tract directly.

Treatment

- **Most spontaneous NCF need surgery** — by diagnosis much parenchyma is lost and inflammation advanced, especially with an obstructing stone or staghorn.
- **Open nephrectomy is preferred**, because about 90% of kidneys are poorly or non-functioning.
- **Conservative options:** heminephrectomy in a duplex system; **functioning kidney with minor parenchymal involvement → stent or nephrostomy may cure.**

Concept — *urocutaneous fistulae persist for the same reasons as all fistulae — epithelialisation, infection, foreign body and distal obstruction — so the vesical and urethral varieties usually close once flow is diverted and obstruction relieved, whereas the nephrocutaneous fistula signals a kidney already destroyed.*

URO-OSTEO-ARTICULAR FISTULAE**Pubovesical Fistula****Aetiopathogenesis and Epidemiology**

- **Pubovesical fistula (PVF)** — rare tract running anteriorly between the **pubic symphysis and bladder, prostate or urethra.**

- **Typical setting: radiotherapy for CaP plus endoscopic outlet procedures** — TURP, photoselective vaporisation of the prostate (PVP), bladder neck incision (BNI), urethral dilatation or incision. **Incidence is rising with multimodal cancer treatment.** Also after posterior urethral reconstruction (sutures near the pubis).
- **Proposed sequence:** irradiated tissue + repeated endoscopy for stenoses → urine leaks into the retropubic space → fistulation into the symphysis.
- **After AUS implantation:** raised outlet resistance increases bladder pressure → rupture of poor-quality irradiated urethra.
- **Series:** the great majority had prior radiation (external beam, brachytherapy or both); a small minority developed PVF after prostatectomy alone; **median interval from radiation to diagnosis about 10 years;** most had a bladder neck endoscopic procedure beforehand.
- **Symptoms: pain on ambulation — the commonest,** irradiated or not; collections, abscess, urosepsis in both; **cutaneous fistulation only in irradiated patients;** in non-irradiated, UTI, haematuria, worsening LUTS.

Diagnosis

- **High index of suspicion: pain on walking or pubic pain after BNI, transurethral prostate surgery, or posterior urethral or bladder neck reconstruction** → investigate for PVF. Imaging usually combined.
- **Nuclear medicine:** sensitive but nonspecific. **CT:** excellent reconstructions; cortical thickening, sclerosis, draining sinuses; earlier than plain films **but inferior to MRI.**
- **MRI is the most sensitive and specific** (and non-invasive): tract with **low central signal and enhancing walls on T1 fat-suppressed post-gadolinium;** parasymphyseal cortical erosion with **symphyseal separation;** collections, abscess, marrow signal change, muscle and tendon inflammation. MRI is the primary diagnostic tool in the great majority.
- **Cystoscopy** shows the tract and outlet dysfunction; cystography shows the fistula; UDS assesses the bladder.
- **CT-guided pubic bone biopsy with cultures** guides antibiotics. Bone cultures are usually positive — commonly **Candida, Enterococcus, Pseudomonas;** concordance with urine culture is inconsistent (**do not rely on urine culture**).

Treatment

- **Multidisciplinary:** urology, orthopaedics, infectious disease. Antibiotics against urosepsis.
- **Conservative care usually fails (75–94% need surgery).** Option for non-septic, previously untreated or frail patients: antimicrobials, drainage of collections, **suprapubic catheter or nephrostomies.**
- **Surgery: remove infected bone, excise the fistula, divert urine.**
 - **Primary repair versus anterior exenteration with diversion** is decided by the criteria in the table below.
 - **Symphysis resected and adjacent rami removed to bleeding bone;** aerobic, anaerobic and fungal cultures; defect filled with a **pedicled rectus abdominis or omental flap.**
 - **4–6 weeks of (usually intravenous) antibiotics** for osteomyelitis.
- **QoL after extirpation:** physical function improves and opioid use falls; mental health scores unchanged.
- **Robotic cystectomy with holmium:yttrium-aluminium-garnet (Ho:YAG) laser debridement of the symphysis:** necrotic cartilage and bone removed (toothed grasper, then laser) **until healthy tissue, preserving the anterior ligaments and the pelvic ring** → high resolution, few redo debridements. Series suggest **pubic bone excision is not mandatory** — infected bone recovers with diversion, debridement and antibiotics.

PVF — PRIMARY REPAIR VERSUS ANTERIOR EXENTERATION

Primary fistula repair	Anterior exenteration + diversion
Functional sphincter	Incompetent sphincter
Normal bladder function, no radiation cystitis	Radiation cystitis or poor bladder
No distal urethral obstruction	Distal obstruction

Concept — PVF is osteomyelitis driven by urine: radiated outlet tissue broken by repeated endoscopy leaks into the retropubic space, so cure requires both removing the infected bone and stopping urine from reaching it — which is why diversion is usually part of the operation.

UROVASCULAR FISTULA**Renovascular Fistula****Arterial Fistula**

- **Artery-to-collecting-system fistulae** (renal pelvis or calyces) are **rare but life threatening**; typically after **transplantation or PCNL**.
- **Mechanisms**: inadvertent procedural injury, and **raised collecting-system pressure causing forniceal rupture**.
- **Arteriocalyceal fistula (ACF)** also after **retrograde intrarenal surgery**, prolonged nephrostomy drainage and trauma; often coexists with pseudoaneurysm or arteriovenous fistula.
- **Angioembolisation is the treatment of choice**; open partial nephrectomy is exceptional.

Ureterovascular Fistula**Aetiopathogenesis and Epidemiology**

- **Ureteroarterial fistula (UAF)** — once exceptional, now increasingly recognised: ageing population, better cancer survival, **multimodal pelvic cancer treatment. A medical emergency**.
- **Primary (15%)**: intrinsic vessel weakness — **aortoiliac aneurysm**, rarely arteriovenous malformation.
- **Secondary (85%)**: fibrosis, inflammation or displacement from **vascular** (grafts, endovascular procedures, surgery), **urological** (diversion, **indwelling stents**, strictures) and other factors (pelvic surgery, cancer, radiation).
- **Main risk factors**: **pelvic surgery, long-term ureteric stents, malignancy, radiotherapy**. Cancers: **cervical, bladder, colorectal**.
- **Site**: **common iliac artery most often, then external iliac**; less often internal iliac or aorta. Ureterovenous fistulae and involvement of colon, small bowel and skin are reported.
- **Latency**: months to more than 10 years after pelvic treatment.
- **Presentation**: **haematuria in essentially all** (often the only sign), possibly intermittent; transient retention, fever or flank pain from clot distending the collecting system; **about 20% haemodynamically unstable**.

Diagnosis

- **Hard to confirm** — imaging insensitive; the aim is exact localisation for treatment; intermittent haematuria makes it harder.
- **Pelvic angiography is the most effective test**: contrast leaks through the defect and fills the ureter. **Sensitivity 25–50%, rising to 60–100% with provocation** — removing catheters or stents during angiography, or thrombolytics.

- **Provocation can precipitate severe haemorrhage** → do it only with **immediate treatment ready**. A negative provocation does not exclude UAF if haematuria persists with risk factors.
- **Cystoscopy: pulsatile blood from one ureteric orifice lateralises**. **Retrograde pyelography** is unreliable — blood prevents the pressure gradient needed.
- **CT angiography**: sensitivity 38–50%; useful for abscess, bowel and vessel anatomy for planning; **usually shows only nonspecific clot and hydronephrosis**, not the wall defect.

Treatment

- **Coordinated urinary and vascular treatment** by radiology, urology and vascular surgery.
- **Endovascular repair is preferred** (less invasive, complex comorbid anatomy). **Stent graft is the main treatment**, embolisation occasionally; internal iliac embolisation is often combined with stent grafting. **Haematuria stopped in all cases**.
- **Complications** over follow-up: rebleeding, stent thrombosis, retroperitoneal abscess, urosepsis, iliac thrombosis, claudication; a few later nephrectomies. **Mortality is high overall, but mostly from the underlying disease** rather than the fistula.
- **Open repair is reserved for** ureteric–intestinal fistula, abscess, or graft infection after endovascular treatment: bypass grafting, proximal arterial ligation, open vessel repair, **nephrectomy if the collecting system is irreversibly damaged**.

Concept — *UAF is the fistula that bleeds rather than leaks: suspect it in any patient with a long-term stent after pelvic surgery and radiation who has haematuria, and remember that the test sensitive enough to find it — provocative angiography — must be done with the endovascular cure ready on the table.*

RENAL FISTULA

Posttrauma

Aetiopathogenesis and Epidemiology

- **Renal injuries are 1–5% of trauma**, often with polytrauma; typically young males; **blunt in about 90%** (vehicle accidents, falls, sport, assault) versus **penetrating** (stab, gunshot — unpredictable and severe).
- **Urinary leak follows 1–7% of renal injuries; up to about a quarter of grade IV** (and similar in grade V); higher (up to 30%) with penetrating trauma.
- **Presentation**: besides instability and other injuries — haematuria, flank pain, sepsis, renal dysfunction.

Diagnosis

- **CT is the gold standard in stable patients**: multiphase from diaphragm to ischial tuberosities — **precontrast, arterial, nephrographic and 10-minute delayed** phases. **Delayed phase may be omitted** if kidneys are normal and there is no perinephric, retroperitoneal, pelvic or perivesical fluid.
- **Ultrasound**: fast, available, cheap, no radiation, good for free fluid, collections, hydronephrosis and follow-up; **excretory CT preferred** where immediately available.
- **MRI**: no advantage; for severe contrast allergy. **Retrograde pyelography** when CT cannot exclude collecting-system injury or **ureteropelvic junction (UPJ) avulsion** — **non-opacification of the distal ureter suggests UPJ injury**.
- **Reimaging**: advisable for severe injuries managed conservatively (bleeding, persistent or new extravasation, vascular malformation); practice varies — some repeat **CT at 36–72 hours for grade III–V**, others only on deterioration. **Always image for fever, flank pain, flank mass or bleeding**. Routine and indicated reimaging find similar complications, all **within the first 3 weeks**.

Treatment

- **Non-operative management is standard** — better renal salvage, fewer complications, fewer explorations and emergency nephrectomies.
- **Most extravasation resolves spontaneously (76–87%). Persistent leak or urinoma → retrograde stent or percutaneous nephrostomy**; percutaneous urinoma drainage is seldom needed.

Postsurgical

Aetiopathogenesis and Epidemiology

- **Partial nephrectomy (PN) is preferred over radical nephrectomy (RN) for T1 tumours** and increasingly for T2 — renal function preserved with equivalent oncological control; larger tumours raise leak and bleeding risk.
- **Urine leak after PN: 0.5–20%.**
- **Risk factors:** collecting-system repair at renorrhaphy, **volume of parenchyma removed**, tumour size and **central location**, operative and warm ischaemia time, early surgeon experience, moderate–severe chronic kidney disease.
- **CT predictors:** higher **RENAL nephrometry score** (Radius, Exophytic/endophytic, Nearness to the collecting system, Anterior/posterior, Location) — **R and N** components, not E; longer ischaemia. On multivariable analysis **only pelvicalyceal contact independently predicted leak**.
- **Robot-assisted PN has a low leak rate** (under 1% in a multicentre series).

Diagnosis

- **Drain output** with drain-versus-serum creatinine; leaks may appear **after drain removal**. Confirm by **multiphase CT**, less often retrograde pyelography.

Treatment

- **Most are managed conservatively with the drain alone**; also ureteric stent, percutaneous drain or nephrostomy.
- **"Active" treatment for persistent leaks** succeeds in nearly all (mean about 11 days to resolution):
 - **Percutaneous:** **cyanoacrylate** (most) or fibrin glue injection; infundibular dilatation plus nephrostomy; laser ablation; CT-guided cryoablation.
 - **Retrograde:** glue injection, or **widening drainage** — double stents, Malecot catheter, exteriorised stent, stent plus endopyelotomy stent, retrograde aspiration system.
 - **Other:** renal artery embolisation, **desmopressin**.
- **Routine drains may add nothing** after robot-assisted PN (stay, complications, renal function), given the low leak rate and spontaneous resolution.

RENAL URINE LEAK — TRAUMA VERSUS PARTIAL NEPHRECTOMY

	Posttrauma	After PN
Frequency	1–7% of injuries; up to about a quarter of grade IV–V	0.5–20%; low after robotic PN
Key predictor	Grade; penetrating mechanism	Pelvicalyceal contact (central tumour)
Diagnosis	Excretory-phase CT	Drain creatinine; multiphase CT
First line	Observe; stent or nephrostomy if persistent or urinoma	Drain alone ; stent or nephrostomy

	Posttrauma	After PN
Refractory	—	Glue (cyanoacrylate), drainage widening, embolisation

Concept — *a renal urine leak, whether from trauma or partial nephrectomy, usually seals if urine is given an easier path down the ureter — hence observation and internal drainage first, and active closure only for the persistent leak.*

VESICourethRAL ANASTOMOTIC FISTULA

Etiopathogenesis and Epidemiology

- **The vesicourethral anastomosis (VUA)** in RP largely determines continence, catheter duration, and the risks of stricture and leakage; minimally invasive surgery allows a more precise watertight anastomosis, shorter catheterisation and faster recovery.
- **Suture techniques: continuous (CS) versus interrupted (IS); in robot-assisted radical prostatectomy (RARP) the continuous Van Velthoven anastomosis prevails.**
- **Radiological leak: 3.2–33%**, depending on definition and catheter duration.

PATIL CLASSIFICATION OF VUA LEAK (CONVENTIONAL CYSTOGRAPHY)

Grade	Definition
I	Extraperitoneal, within 6 cm of the VUA
II	Extraperitoneal, extending to the sidewall ≥6 cm from the VUA
III	Intraperitoneal

- **Patil** pioneered quantifiable leak definitions: with pre-removal cystography, a significant leak meant leaving the catheter and repeating the cystogram at 7 days; a few needed pelvic drains, one had urinoma-related ileus, and some later developed bladder neck contracture.

Diagnosis

- **Pre-removal cystography** is used, but **leak extent may not correlate with outcome**. Prolonged catheterisation is linked to **poorer function and more complaints**; conversely leakage is linked to **retention, anastomotic stricture and voiding complaints**.
- **Risk-adapted cystography** for men with **large prostates and preoperative LUTS**, with longer catheterisation if a leak is shown.
- **Clinically significant leak**: drain fluid with raised creatinine, or substantial extravasation on a **day-7 cystogram** that extends the planned catheter period.

Treatment

- **Extended catheterisation** is the mainstay; **gentle catheter traction** is common.
- **Correct catheter position**: for a non-fenestrated catheter — radiological repositioning, balloon deflation, external fixation.
- **Drain manoeuvres**: switch active to passive suction, pull back, withdraw.
- **Persistent leak** → **ureteric stents and percutaneous nephrostomies** before surgery.

- **Reoperation (last resort)** depends on urine volume and the original route; after a **transperitoneal RP**, leak causes **uoperitoneum and ileus** (though significant leaks may lack them). Options range from partial repair to complete re-anastomosis.

Concept — *anastomotic leak is managed by stepwise improvement of drainage — catheter, then drain, then upper-tract diversion — with reoperation reserved for failure, because most leaks seal once urine preferentially exits through the catheter.*

URINARY FISTULA AFTER GENDER-AFFIRMING SURGERY

- **Rising numbers** mean urologists meet unfamiliar anatomy; genital reconstruction is **multistage and complex**, so complications are common.

Female to Male Gender-Affirming Surgery

- **Urethral lengthening is the major challenge** — strictures, obstruction and fistulae.
- **After phalloplasty and metoidioplasty, stricture and fistula are both common**; fistula relatively commoner after metoidioplasty; both typically appear **within about 3 months**. Most need revision, after which most void from the tip of the phallus.
- **Colpectomy (primary) roughly halves urethral fistula rates** after single-stage urethral lengthening.
- **Management**: small fistulae often close with **prolonged catheterisation**; persistent → excision and primary repair or complex regrafting.

Male to Female Gender-Affirming Surgery

- **Urethrovaginal (urethrovaginal) fistula is rare** — continuous incontinence and recurrent UTI. **Conservative first** (suprapubic diversion); failure → **formal urethroplasty with omentum or other interposition** if complex. Rates are low after both **penile inversion and penoscrotal vaginoplasty**.
- **Vesicovaginal fistula** (about 0.9%) — usually from intraoperative bladder injury; conservative first, then **perineal or transabdominal repair** with interposition in complex cases.

FISTULA AFTER GENDER-AFFIRMING SURGERY

	Transmasculine (phalloplasty, metoidioplasty)	Transfeminine (vaginoplasty)
Frequency	Common , with stricture	Rare
Typical fistula	Neourethral (urethrocutaneous)	Urethrovaginal; vesicovaginal
Risk reduction	Primary colpectomy	Avoid bladder injury
Management	Catheter → excision and repair or regrafting	Suprapubic diversion → urethroplasty or repair with interposition

Concept — *fistula risk after gender-affirming surgery follows urethral construction: the transmasculine neourethra is long, newly built and poorly supported, so fistula is common, whereas the transfeminine urethra is native and shortened, so fistula is rare and usually reflects a direct intraoperative injury.*